

12 Thermal Control

12.1 Introduction

Thermal control, whether of an instrument or spacecraft, is required to maintain the temperatures of components and subsystems within different specified ranges. As a result, the thermal design must satisfy a range of temperature requirements for a variety of dynamic thermal environments. This is achieved by developing the thermal control system through design, analysis, and test with constraints to minimize cost, mass, power, and volume. Thermal design is based on managing heat flux within the spacecraft, to the spacecraft, and from the spacecraft. This must be accomplished by satisfying the different temperature requirements for all phases of the mission and operating modes including prelaunch, launch, early flight operations, and possibly several mission scenarios from the beginning of life (BOL) to the end of life (EOL) of the mission. The different phases and environments to which an instrument or spacecraft is exposed are illustrated in Table 12.1.

The specifications for the thermal control systems are generally defined in terms of one or more of the following for each component: 1) operating temperature range, 2) non-operating or turn-on temperature range, 3) survival temperature range, and 4) thermal deformation upper bound. *Operating temperature range* is the range over which the device is designed to operate safely and meet its performance and reliability specifications. *Non-operating/turn-on temperature range* is the temperature range within which a device can be powered on and will meet its performance and reliability specifications when within its operating temperature range. *Survival temperature range* is the temperature range to which the device can be exposed without damage when unpowered. The upper bound to the thermal deformation is the absolute differential limit that a component can deform due to heating or cooling and still meet its spatial performance criteria. This is generally imposed on precision pointing devices such as optical systems. In most cases, the range of operating temperatures can be quite wide, on the order of 50–100°C, although in some cases, in devices requiring precise alignments, the permitted temperature range could be as small as 1°C. Table 12.2 illustrates some example temperature tolerances.

Criteria by which to evaluate a thermal control system are identified in Table 12.3 adapted from Gilmore (2001).

Complementary references for thermal control of space systems include Gilmore (2001), Moran et al. (2003), Pisacane (2005), Karam (1998), and Tribble (2003).

Table 12.1 Mission phases and environments

Mission phase	Typical time periods	Environment
Development	Months to years	Clean room Temperature varies by <20% Relative humidity <40%
Transport	Days	Hermetically sealed with nitrogen Exposed to shock, vibration, and temperature variations
Launch	≈0.5 h	Exposed to shock, vibration, and temperature variations Aerodynamic heating of heat shield up to 500°C Solar illumination, planetary albedo, and planetary thermal radiation
Transfer operations	Hours to days	Variable solar illumination, planetary albedo, and planetary thermal radiation Vacuum Various orientations Various operating modes Subsystem energy dissipation Heat from propulsion system
Operations	Months to years	Variable solar illumination, planetary albedo, and planetary thermal radiation Vacuum Various orientations Stable operating modes Subsystem energy dissipation Heat from propulsion system

12.2 Spacecraft Thermal Environments

In space, the thermal environment of a spacecraft includes 1) direct solar radiation, 2) albedo radiation, 3) thermal energy radiated by nearby celestial bodies, 4) deep space at 4 K, and 5) onboard heat sources (radioisotopes, propellants, electrical dissipation). These are illustrated in Fig. 12.1 and are discussed in turn.

Table 12.2 Subsystem temperature tolerances

Component	Operating temperatures, °C	Non-operating/turn-on temperature, °C	Survival temperature, °C
Typical electronics	−15 to +45	−30 to +50	−30 to +60
Batteries	0 to +25	−10 to +30	−10 to +30
Infrared detectors	−269 to −173	−269 to −173	−269 to +35
Fuel (hydrazine)	+10 to +50	120	+5 to +110
Solar cells	−100 to +120	−100 to +120	−100 to +120
Gyros/reaction wheels	0 to +40	−5 to +45	−10 to +50
Optical systems	21±1	21±1	21±1
Antennas	−90 to +100	−90 to +100	−90 to +100

Table 12.3 Thermal control system evaluation criteria

Criteria
Satisfies specifications
Adequate margin over specifications
Minimal single point failures
Reliable
Robust
Has flight heritage
Predictable by thermal analysis
Verifiable by ground test
Meets lifetime requirement
Can survive launch environment
Performance not degraded by space environment
Components satisfy NASA outgassing limits
Imposes minimum amount of operational restrictions
Allows for proper venting and outgassing
Is fault tolerant

Source: Adapted from Gilmore (2001).

Direct solar electromagnetic radiation was described in Chapter 3 but is repeated here for completeness. The solar irradiance of the Earth at 1 AU from the sun, or *solar constant*, as discussed in Chapter 3, has the accepted mean value of 1366.1 W m^{-2} and increases by 3.428% at perihelion and decreases by 3.260% at aphelion. Solar electromagnetic radiation is a function of wavelength, with its spectral irradiance illustrated in Fig. 3.12. The heat flux density at any other distance from the sun is given by

$$\dot{q} = S_e \left(\frac{a_{\text{Earth}}}{r_s} \right)^2 = \frac{P_s}{4\pi r_s^2} \quad (12.1)$$

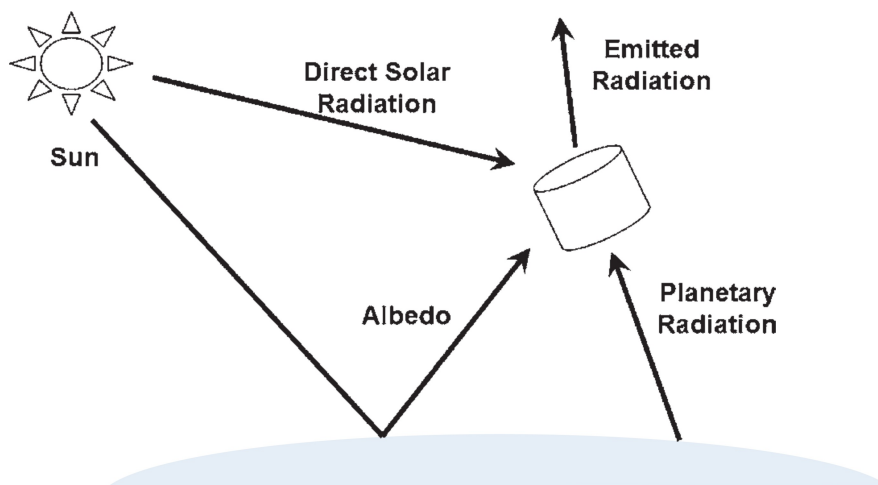


Fig. 12.1 Energy exchange with the environment.

where

- a_{Earth} = average distance of Earth from sun, 1 AU, m
- P = total power of the sun, W
- $\dot{q}$ = heat flux density, W m^{-2}
- r_s = distance from sun, m
- S_e = solar constant for the Earth, W m^{-2}

The mean solar constants for the planets follow from Eq. (12.1) and are illustrated in Table 3.2.

Albedo, more specifically the Bond albedo as discussed in Chapter 3, is defined as the amount of electromagnetic radiation reflected from a surface to the amount of energy incident on it. Bond albedos for the planets averaged over their surfaces are given in Table 3.5. However, albedo can vary significantly over a celestial body. For example, on Earth the Bond albedo for snow is 0.75–0.85, for land 0.05–0.45, for water 0.03–0.20, and for clouds 0.10–0.80. The spectral content of the albedo is generally assumed to be that of the solar radiation.

The thermal energy radiated by a celestial body is determined primarily by the amount of solar radiation incident on it. The blackbody temperature and radiated thermal power density (radiant exitance) of the celestial bodies in the solar system are given in Table 3.6.

12.3 Heat Convection

Convection is the transfer of heat between a surface and a moving fluid. The fluid motion may be caused naturally in a gravitational field by temperature differences or forced by mechanical means such as a blower. Convection is described quantitatively by

$$\dot{q} = h\Delta T = h |T_{\text{fluid}} - T_{\text{surface}}| \quad (12.2)$$

where

- h = convective heat transfer coefficient, $\text{W m}^{-2} \text{K}^{-1}$
- $\dot{q}$ = heat flux density, W m^{-2}
- $T_{\text{fluid}}, T_{\text{surface}}$ = temperatures, K
- ΔT = temperature difference, K

The convective heat transfer coefficient depends on the physical properties of the gas or fluid. For most gases it is between 1 and 20 $\text{W m}^{-2} \text{K}^{-1}$ while typical values for fluids are between 100 and 1000 $\text{W m}^{-2} \text{K}^{-1}$. Forced convection is generally used to control the temperature in a spacecraft and launch vehicle while both are on the launch pad. Convection is also important in human space exploration missions where atmospheres are maintained in the spacecraft and in the space suit during extravehicular activities. In the absence of gravity, it is necessary to depend on forced convection to achieve the desired uniformity of temperature. Most robotic spacecraft systems operate in the vacuum of space

where heat transfer by convection is negligible and is ignored. However, selected electronic packages may be hermetically sealed, for which convection could be important for their operation. Pumped coolant loops and loop heat pipes are used in spacecraft and space instrumentation, as well as discussed in Section 12.6.10.

12.4 Heat Conduction

12.4.1 Heat Conduction Equation

Conduction is heat transfer from one part of a body to another under the influence of a temperature gradient without appreciable displacement of particles. Thus, it primarily involves the transfer of heat or energy from one molecule to the next within solids, but can also occur in liquids and gases. Conduction was described quantitatively by Fourier in 1822 by

$$\dot{\mathbf{q}} = -k \nabla T \quad (12.3)$$

which for a single spatial dimension reduces to

$$\dot{q} = -k \frac{dT}{ds} \quad (12.4)$$

where

k = thermal conductivity, $\text{W m}^{-1} \text{K}^{-1}$

$\dot{\mathbf{q}}$ = heat flux density vector, W m^{-2}

$\dot{q}$ = heat flux density, W m^{-2}

s = displacement, m

T = temperature, K

dT/ds = spatial temperature gradient, K m^{-1}

Examples of thermal conductivity are given in Table 12.4.

Thermal conductance G , $\text{W m}^{-2} \text{K}^{-1}$, is defined as the heat flux density divided by the temperature difference ΔT :

$$G = \frac{\dot{q}}{\Delta T} \quad (12.5)$$

Conduction can be an important source of transfer of heat within an instrument or spacecraft, especially the conduction of heat to exterior surfaces, where it can then be radiated into space. The heat equation for conduction follows from the conservation of energy for a small element as

$$\rho c_p \frac{\partial T(r, t)}{\partial t} - \nabla \cdot (k \nabla T(r, t)) = \dot{q}_v \quad (12.6)$$

where

c_p = specific heat capacity, $\text{J kg}^{-1} \text{K}^{-1}$

k = thermal conductivity, $\text{W m}^{-1} \text{K}^{-1}$

Table 12.4 Example thermal conductivity coefficients

Material	Thermal conductivity, W m ⁻¹ K ⁻¹
Air	0.024
Aluminum	250
Aluminum oxide	30
Beryllium	218
Brass	109
Carbon	1.7
Carbon steel	54
Epoxy	0.35
Fiberglass	0.04
Glass	0.15
Gold	310
Helium	0.142
Iron	80
Kerosene	0.15
Nickel	91
Polyethylene	0.46
Silver	429
Stainless steel	16
Tin	67
Water	0.58
Zinc	116

$\dot{q}_v$ = volumetric heat generation, W m⁻³

r = denotes a position in space, m

ρ = density, kg m⁻³

ρc_p = volumetric heat capacity, J m⁻³ K⁻¹

If the thermal conductivity is spatially constant, then Eq. (12.6) can be expressed as

$$\frac{\partial T(r, t)}{\partial t} - \kappa \nabla^2 T(r, t) = \frac{\dot{q}_v}{\rho c_p} \quad (12.7)$$

where the thermal diffusivity κ is defined by the ratio of the thermal diffusivity to the volumetric heat capacity:

$$\kappa \equiv \frac{k}{\rho c_p} \quad (12.8)$$

where

κ = thermal diffusivity, m² s⁻¹, often given in cm² s⁻¹

Table 12.5 gives the thermal diffusivity of selected materials.

Table 12.5 Example thermal diffusivities of selected materials

Material	Thermal diffusivity, cm ² s ⁻¹
Air	0.19
Aluminum	0.999
Aluminum oxide	0.088
Ceramic	0.02
Copper	1.13
Glass	0.0043
Gold	1.27
Nylon	0.0013
PTFE ^a	0.0011
Silver	1.7004
Stainless steel	0.0405
Water	0.0014

^aPTFE = polytetrafluoroethylene.

12.4.2 Exact Solutions

Two examples are presented to illustrate the determination of an exact solution to the heat equation for conduction.

Example 12.1

Problem: Determine the steady-state distribution of temperature in an infinitely long plate with thickness L and temperature T_i on one side and T_o on the other side.

Solution: For the steady-state heat flux distribution along a single dimension without volumetric heat generation, the heat equation, Eq. (12.7), reduces to

$$\frac{d^2 T}{ds^2} = 0 \quad (12.9)$$

Integration gives

$$T = c_1 s + c_2 \quad (12.10)$$

Substituting the initial conditions $T = T_i$ at $s = 0$ and $T = T_o$ at $s = L$ gives the solution to be

$$T = \frac{T_o - T_i}{L} s + T_i \quad (12.11)$$

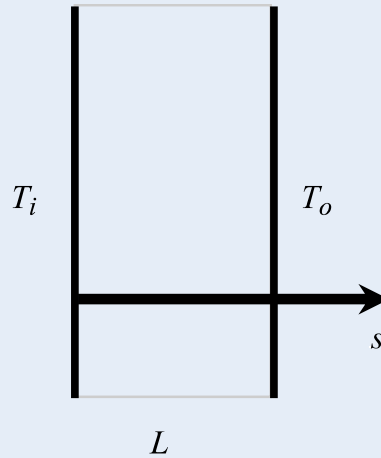


Fig. 12.2 Geometry for Example 12.1.

which indicates a linear temperature variation with distance. From Fourier's law, Eq. (12.4), the heat flux density $\dot{q}$ is

$$\dot{q} = -k \frac{dT}{ds} = -k \frac{T_o - T_i}{L} \quad (12.12)$$

Example 12.2

Problem: Determine the general transient solution to the heat conduction equation in one dimension.

Solution: The one-dimensional heat conduction equation without volumetric heat generation and constant thermal diffusivity follows from Eq. (12.7) as

$$\frac{\partial T}{\partial t} - \kappa \frac{\partial^2 T}{\partial s^2} = 0 \quad (12.13)$$

Assume that the solution is separable where

$$T(s, t) = T_s(s)T_t(t) \quad (12.14)$$

which upon substitution into Eq. (12.13) gives

$$T_s \frac{\partial T_t}{\partial t} - \kappa T_t \frac{\partial^2 T_s}{\partial s^2} = 0$$

or

$$\frac{1}{\kappa T_t} \frac{\partial T_t}{\partial t} = \frac{1}{T_s} \frac{\partial^2 T_s}{\partial s^2} = -\frac{1}{\alpha^2} \quad (12.15)$$

where each side must be equal to a constant and where $-\alpha^{-2}$ is the separation constant. The solution to

$$\frac{1}{T_t} \frac{dT_t}{dt} = -\frac{\kappa}{\alpha^2} \quad (12.16)$$

is

$$T_t = C_t e^{-\kappa t / \alpha^2}$$

The solution to

$$\frac{d^2 T_s}{ds^2} + \frac{1}{\alpha^2} T_s = 0 \quad (12.17)$$

is

$$T_s = C_{s1} \cos\left(\frac{s}{\alpha}\right) + C_{s2} \sin\left(\frac{s}{\alpha}\right) \quad (12.18)$$

The solution is then

$$T(s, t) = T_t(t) T_s(s) = e^{-\kappa t / \alpha^2} \left[C_1 \cos\left(\frac{s}{\alpha}\right) + C_2 \sin\left(\frac{s}{\alpha}\right) \right] \quad (12.19)$$

where the constants α , C_1 , and C_2 must be satisfied by the initial conditions and the boundary conditions.

12.4.3 Finite Difference Method

Exact analytical solutions to the nonlinear heat conduction equation, Eq. (12.7), are useful in theory but generally not possible in practice. Consequently, several approximate techniques have been developed. One of the more popular and useful is the finite difference approach, of which there are several variations. The finite difference solution of the heat equation is based on discretizing space by an integer number of spatially distributed points and time by an integer number of discrete time steps. As a result, the differentials may be replaced by finite differences, resulting in systems of algebraic equations. In a two-dimensional medium, the heat equation, Eq. (12.7), reduces to

$$\frac{\partial T}{\partial t} = \kappa \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) + \frac{\dot{q}_v(x, y, t)}{\rho c_p} \quad (12.20)$$

Figure 12.3 illustrates the temperatures at discrete grid points for a two-dimensional surface.

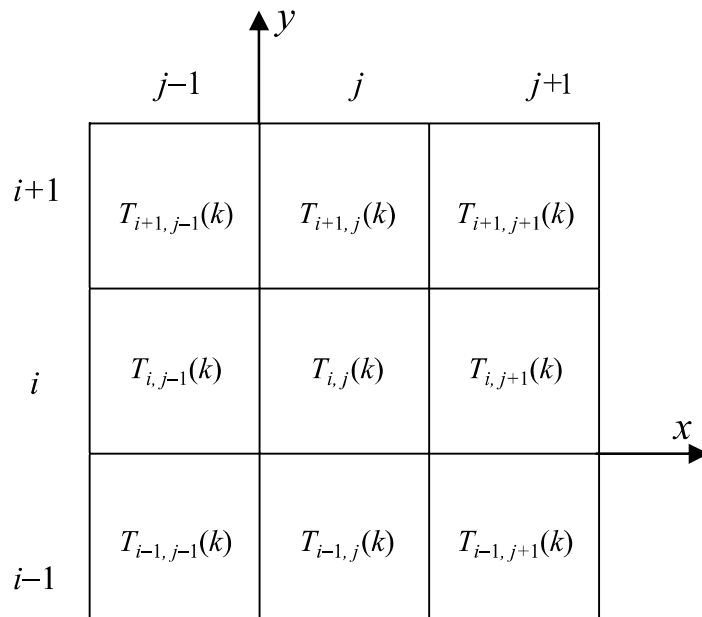


Fig. 12.3 Two-dimensional array of discrete elements at time step k

From Fig. 12.3, the time differential at the i, j node can be approximated by the forward difference:

$$\frac{\partial T_{i,j}(k)}{\partial t} \approx \frac{T_{i,j}(k+1) - T_{i,j}(k)}{\Delta t} \quad (12.21)$$

and the spatial differentials can be approximated by the forward differences:

$$\frac{\partial^2 T_{i,j}(k)}{\partial x^2} = \frac{\frac{\partial T_{i,j+1}(k)}{\partial x} - \frac{\partial T_{i,j}(k)}{\partial x}}{\Delta x} = \frac{T_{i,j+1}(k) - 2T_{i,j}(k) + T_{i,j-1}(k)}{(\Delta x)^2} \quad (12.22)$$

$$\frac{\partial^2 T_{i,j}(k)}{\partial y^2} = \frac{\frac{\partial T_{i+1,j}(k)}{\partial y} - \frac{\partial T_{i,j}(k)}{\partial y}}{\Delta y} = \frac{T_{i+1,j}(k) - 2T_{i,j}(k) + T_{i-1,j}(k)}{(\Delta y)^2} \quad (12.23)$$

Substituting Eqs. (12.21–12.23) into Eq. (12.20) gives

$$\begin{aligned} \frac{T_{i,j}(k+1) - T_{i,j}(k)}{\Delta t} = & \kappa \frac{T_{i,j+1}(k) - 2T_{i,j}(k) + T_{i,j-1}(k)}{(\Delta x)^2} \\ & + \kappa \frac{T_{i+1,j}(k) - 2T_{i,j}(k) + T_{i-1,j}(k)}{(\Delta y)^2} + \frac{\dot{q}_{v,i,j}(k)}{\rho c_p} \end{aligned}$$

$$T_{i,j}(k+1) = T_{i,j}(k) + \frac{\kappa \Delta t}{(\Delta x)^2} (T_{i,j+1}(k) - 2T_{i,j}(k) + T_{i,j-1}(k)) + \frac{\kappa \Delta t}{(\Delta y)^2} (T_{i+1,j}(k) - 2T_{i,j}(k) + T_{i-1,j}(k)) + \frac{\dot{q}_{v,i,j}(k) \Delta t}{\rho c_p} \quad (12.24)$$

This equation states that the new temperature at a node is the old temperature at that node plus the weighted average of the differences between the old temperature at that node and the temperatures of the four surrounding nodes plus the volumetric heat generation. The typical solution involves the specification of the initial conditions on temperature, the characteristics of the medium, and the volumetric heat production $\dot{q}_v$ with the values for Δx , Δy , and Δt selected to assure convergence of the solution. Stable solutions are obtained when

$$\frac{\kappa \Delta t}{(\Delta x)^2} = \frac{\kappa \Delta t}{(\Delta y)^2} < \frac{1}{2} \quad \text{for the one-dimensional case} \quad (12.25)$$

$$\frac{\kappa \Delta t}{(\Delta x)^2} = \frac{\kappa \Delta t}{(\Delta y)^2} < \frac{1}{4} \quad \text{for the two-dimensional case} \quad (12.26)$$

$$\frac{\kappa \Delta t}{(\Delta x)^2} = \frac{\kappa \Delta t}{(\Delta y)^2} < \frac{1}{6} \quad \text{for the three-dimensional case} \quad (12.27)$$

Example 12.3

Problem: Consider an insulated rod in which the temperature is initially at 0°C , as illustrated in Fig. 12.4. Suppose the temperature at one end is held fixed at 0°C and at the other end is suddenly heated to 100°C . Determine the temperature distribution.

Solution: The schematic for the rod is as shown with the temperatures T_2 and T_3 to be determined.

Equation (12.24) gives for the one-dimensional case

$$T_2(k+1) = T_2(k) + \frac{\kappa \Delta t}{(\Delta x)^2} (T_3(k) - 2T_2(k) + T_1(k)) \quad (12.28)$$

$$T_3(k+1) = T_3(k) + \frac{\kappa \Delta t}{(\Delta x)^2} (T_4(k) - 2T_3(k) + T_2(k)) \quad (12.29)$$

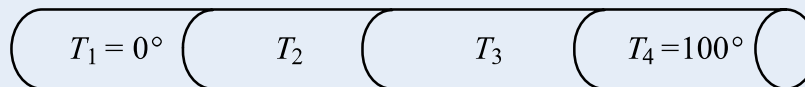


Fig. 12.4 Geometry for example.

Table 12.6 Iterative solutions

Time step	T1	T2	T3	T4
1	0	0	0	100
2	0	0	25.0000	100
3	0	6.2500	37.5000	100
4	0	12.5000	45.3125	100
5	0	17.5781	50.7813	100
10	0	29.5792	62.9124	100
11	0	30.5177	63.8510	100
20	0	33.1219	66.4553	100
21	0	33.1748	66.5081	100
30	0	33.3214	66.6548	100
31	0	33.3244	66.6577	100
40	0	33.3327	66.6660	100
41	0	33.3328	66.6662	100
50	0	33.3333	66.6666	100
51	0	33.3333	66.6666	100

For $\kappa\Delta t/(\Delta x)^2 = 0.25$, the iterative solutions for the temperatures follow where it can be seen that the temperatures converge to $T_2 = 33.3333^\circ\text{C}$ and $T_3 = 66.6666^\circ\text{C}$ as would be anticipated; see Table 12.6.

12.5 Heat Radiation

12.5.1 Fundamentals

The fundamentals of electromagnetic radiation, the transfer of energy by electromagnetic waves, are discussed in Chapter 3 and are summarized here for completeness. *Planck's law of blackbody radiation* for the spectral radiance $L_\lambda(T, \lambda)$ or $L_\nu(T, \nu)$ is given by Eq. (3.5) as

$$L_\nu(T, \nu) = \frac{2h\nu^3}{c^2 \left(\exp \left[\frac{h\nu}{kT} \right] - 1 \right)} \quad (3.5')$$

$$L_\lambda(T, \lambda) = \frac{2hc^2}{\lambda^5 \left(\exp \left[\frac{hc}{\lambda kT} \right] - 1 \right)}$$

where

c = speed of light, m s^{-1}

h = Planck's constant, J s

k = Boltzmann's constant, J K^{-1}

$L_\nu(\nu)$ = spectral radiance, $\text{W m}^{-2} \text{sr}^{-1} \text{Hz}^{-1}$

$L_\lambda(\lambda)$ = spectral radiance, $\text{W m}^{-3} \text{sr}^{-1}$, $\text{W m}^{-2} \text{sr}^{-1} \mu\text{m}^{-1}$
 T = absolute temperature, K
 λ = wavelength $\nu\lambda = c$, m
 ν = frequency, Hz

Figure 3.11 illustrates the spectral radiance for three temperatures where $T = 5900 \text{ K}$ is the approximate blackbody temperature of the sun and $T = 290 \text{ K}$ is the blackbody temperature of the Earth. Figure 3.12 gives the spectral irradiance of the extraterrestrial spectrum reference model.

Spectral emittance ε_λ is a surface property of an object that depends on its surface finish, coating thickness, shape of the object, and wavelength of the emitted radiation. If the emittance of a body is independent of frequency, it is deemed a *gray body* or *gray surface*. *Spectral absorptance* α_λ is defined as the fraction of the incident irradiance absorbed by an object at a particular wavelength. *Spectral reflectance* ρ_λ is the fraction of radiant flux that is reflected by an object at a particular wavelength. *Spectral transmittance* τ_λ is defined as the fraction of the incident radiant energy that, having entered a layer of absorbing material, reaches its further boundary at a particular wavelength. If scattering and luminescence can be neglected, the sum of the spectral transmittance, spectral absorptance, and spectral reflectance is equal to one, where

$$\tau_\lambda + \alpha_\lambda + \rho_\lambda = 1 \quad (3.7')$$

as illustrated in Fig. 3.13.

The *Stefan–Boltzmann law*, discussed in Chapter 3, is obtained by integrating the spectral radiance, Eq. (3.5), for a perfectly diffuse nonblackbody and gives the spectral radiance exitance $M_\lambda(T, \lambda)$ as

$$M_\lambda(T, \lambda) = \dot{q}(T, \lambda) = \pi \varepsilon_\lambda L_\lambda(T, \lambda) \quad (12.30)$$

Assuming that the spectral radiance is independent of wavelength yields Eq. (3.27) for the radiance exitance $M(T)$:

$$M(T) = \dot{q}(T) = \frac{2\pi^5 k^4}{15h^3 c^2} \varepsilon T^4 = \varepsilon \sigma T^4 \quad (12.31)$$

where

$\dot{q}(\lambda, T)$, $M(\lambda, T)$ = radiant exitance or heat flux density, W m^{-2}
 L_λ = spectral radiance given by Planck's law
 M_λ = spectral radiant exitance, W m^{-3} or $\text{W m}^{-2} \mu\text{m}^{-1}$
 T = temperature, K
 ε_λ = spectral emittance, dimensionless between 0 and 1
 σ = Stefan–Boltzmann constant, $\sigma \equiv 2\pi^5 k^4 / 15h^3 c^2 \text{ W m}^{-2} \text{ K}^{-4}$

The wavelength where the spectral radiance is a maximum, $\lambda_{\max}$, is given by *Wien's displacement law*, obtained by differentiating Planck's equation with respect to wavelength, setting the derivative of the spectral radiance to zero, and

solving for wavelength. The resulting equation cannot be solved in closed form but can be approximated by Eq. (3.28):

$$\lambda_{\max} T = C \quad (12.32)$$

where

$C = \text{constant}, 2.8977685 \times 10^{-3}, \text{ m K}$

$T = \text{absolute temperature of the body, K}$

$\lambda_{\max} = \text{radiation wavelength with maximum intensity, m}$

12.5.2 Kirchhoff's Law

Kirchhoff's law states that the spectral absorptance α_λ and the spectral emittance ε_λ at a given wavelength are equal:

$$\alpha_\lambda = \varepsilon_\lambda \quad (12.33)$$

This can be argued on the fact that bodies with different surface properties in a chamber will reach the chamber temperature at equilibrium. It can also be argued mathematically by considering the heat transfer between two parallel infinite diffuse surfaces, as illustrated in Fig. 12.5, in which the sides not facing each other are insulated. Surface 1, assumed to be a blackbody, will emit power according to the Stefan–Boltzmann law, Eq. (12.30), where the spectral radiant exitance $M_{\lambda 1}(T_1, \lambda_1)$ is

$$M_{\lambda 1}(T_1, \lambda_1) = \dot{q}_{1 \rightarrow 2}(T_1, \lambda_1) = \pi \varepsilon_{\lambda 1} L_{\lambda 1}(T_1, \lambda_1) \quad (12.34)$$

where the subscript 1 refers to side 1, and

$L_{\lambda 1}(T_1) = \text{spectral radiance of surface 1, } \text{W m}^{-3} \text{ sr}^{-1}, \text{W m}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1}$

$M_{\lambda 1}, \dot{q}_{1 \rightarrow 2}(\lambda, T_1) = \text{spectral radiant exitance or heat flux density incident on 2, } \text{W m}^{-2}$

$T_1 = \text{temperature of surface 1}$

$\varepsilon_{\lambda, 1} = 1, \text{ emittance of surface 1}$

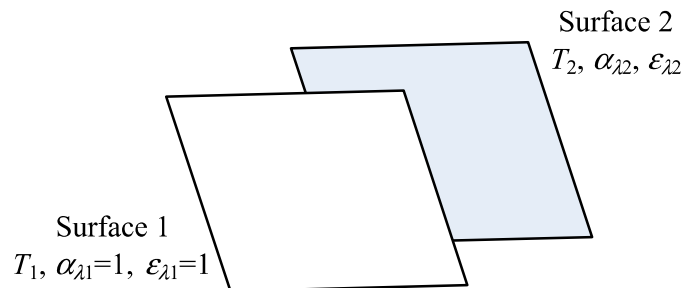


Fig. 12.5 Heat transfer between two infinite diffuse surfaces.

The heat flux density or the spectral radiant exitance $M_{\lambda_2}(T_2)$ emitted from surface 2 to surface 1 according to the Stefan–Boltzmann law is

$$M_{\lambda_2}(T_2) = \dot{q}_{2 \rightarrow 1}(T_2, \lambda_2) = \pi \varepsilon_{\lambda_2} L_{\lambda_2}(T_2, \lambda_2) \quad (12.35)$$

where the subscript 2 refers to side 2, and

$$\begin{aligned} L_{\lambda_2}(T_2) &= \text{spectral radiance of surface 2, } \text{W m}^{-3} \text{ sr}^{-1}, \text{W m}^{-2} \text{ sr}^{-1} \mu\text{m}^{-1} \\ M_2, \dot{q}_{2 \rightarrow 1}(\lambda, T_2) &= \text{radiant exitance or heat flux density incident on 1, } \text{W m}^{-2} \\ \varepsilon_{\lambda, 2} &= \text{emittance of surface 2} \\ T_2 &= \text{temperature of surface 2} \end{aligned}$$

If the heat transfer is in equilibrium at the same temperature, $T = T_1 = T_2$, the heat flux density absorbed by surface 2 from surface 1 must be equal to the heat flux density absorbed by surface 1 from surface 2, so that the spectral radiances are equal:

$$L_{\lambda_1}(T, \lambda_1) = L_{\lambda_2}(T, \lambda_2) \quad (12.36)$$

so that

$$\begin{aligned} \alpha_{\lambda_2} \dot{q}_{1 \rightarrow 2}(T_1, \lambda_1) &= \alpha_{\lambda_1} \dot{q}_{2 \rightarrow 1}(T_2, \lambda_2) \\ \pi \alpha_{\lambda_2} \varepsilon_{\lambda_1} L_{\lambda_1}(T, \lambda_1) &= \pi \alpha_{\lambda_1} \varepsilon_{\lambda_2} L_{\lambda_2}(T, \lambda_2) \\ \alpha_{\lambda_2} &= \varepsilon_{\lambda_2} \end{aligned} \quad (12.37)$$

where

$$\begin{aligned} \alpha_{\lambda_1} &= 1, \text{ spectral absorptance of surface 1} \\ \alpha_{\lambda_2} &= \text{spectral absorptance of surface 2} \\ \varepsilon_{\lambda_1} &= 1, \text{ spectral emittance of surface 1} \\ \varepsilon_{\lambda_2} &= \text{spectral emittance of surface 2} \end{aligned}$$

Thus, the absorptance and emittance at the same wavelength are equal but can be a function of the temperature. It follows by inspection from Eq. (12.36) that when integrated over all wavelengths,

$$\alpha = \varepsilon \quad (12.38)$$

12.5.3 Surface Thermal Properties

Emittance and absorptance of a real body may be a function of the orientation with respect to the normal to the surface. When this is the case, the hemispherical spectral emittance $\varepsilon_{H,\lambda}$ and hemispherical spectral absorptance $\alpha_{H,\lambda}$ are defined by

$$\varepsilon_{H,\lambda} = \frac{1}{\pi} \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} \varepsilon_{\lambda}(\theta) \sin \theta \cos \theta \, d\theta \, d\phi \quad (12.39)$$

$$\alpha_{H,\lambda} = \frac{1}{\pi} \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} \alpha_{\lambda}(\theta) \sin \theta \cos \theta \, d\theta \, d\phi \quad (12.40)$$

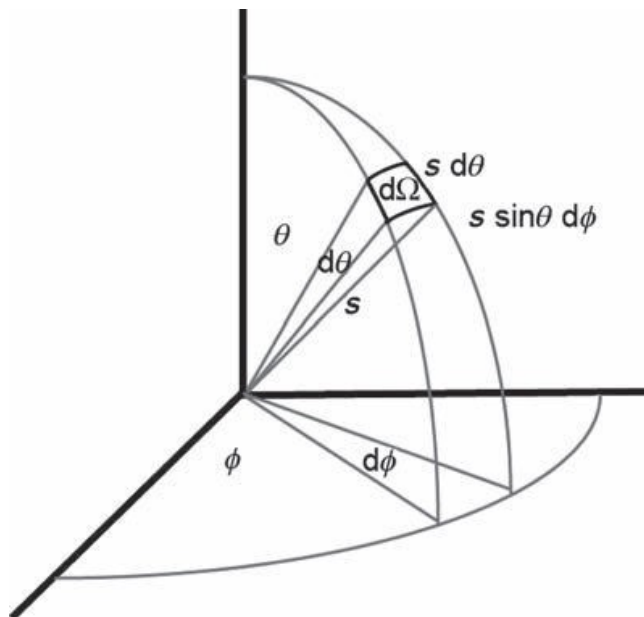


Fig. 12.6 Spherical geometry.

where the geometry is given in Fig. 12.6.

The emittance, absorptance, reflectivity, and transmissibility of a body at a given temperature over a specified range of wavelengths is given by

$$\varepsilon = \frac{\int \varepsilon_{H,\lambda} M_\lambda d\lambda}{\int M_\lambda d\lambda} \quad (12.41)$$

$$\alpha = \frac{\int \alpha_{H,\lambda} E_\lambda d\lambda}{\int E_\lambda d\lambda} \quad (12.42)$$

$$\rho = \frac{\int \rho_{H,\lambda} E_\lambda d\lambda}{\int E_\lambda d\lambda} \quad (12.43)$$

$$\tau = \frac{\int \tau_{H,\lambda} E_\lambda d\lambda}{\int E_\lambda d\lambda} \quad (12.44)$$

Similar to Eq. (3.7'), it follows that

$$\tau + \alpha + \rho = 1 \quad (12.45)$$

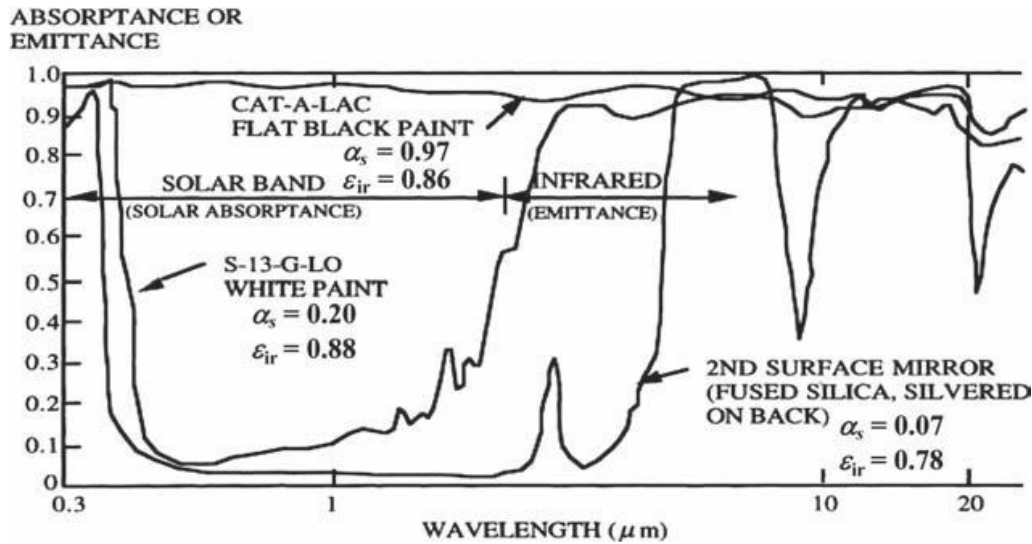


Fig. 12.7 Sample emittance and absorbance as a function of wavelength. [From Silverman (1995); courtesy of NASA.]

The emittance and absorbance of black paint, white paint, and fused silica with the backside silvered is illustrated in Fig. 12.7 as a function of wavelength. It should be noted that the emittance and absorbance are nearly constant but distinctly different over large portions of the solar and infrared bands. Since the emittance and absorbance are equal, it is by convention to call the average hemispherical absorbance and emittance over the solar band the solar absorbance denoted by α_s . Likewise, it is common practice to call the average emittance and absorbance over the infrared band the infrared emittance denoted by ϵ_{ir} . The hemispherical solar absorbance and infrared emittance for the black paint, white paint, and mirror are also given in Fig. 12.7.

Table 12.7 gives the solar (hemispherical) absorbance and infrared (hemispherical) emittance for selected surfaces.

12.5.4 Heat Transfer Between Surfaces

The radiated heat flux exchange over all wavelengths between two differential areas dA_i and dA_j , separated by a distance r , as illustrated in Fig. 12.8, is given by

$$dE_j dA_j = L_i \cos \beta_i d\Omega_j dA_i = \frac{L_i \cos \beta_i \cos \beta_j}{r^2} dA_i dA_j \quad (12.46)$$

where

dA_i, dA_j = differential areas, m^2

E_j = irradiance or heat flux density incident on dA_j from dA_i , $W m^{-2}$

L_i = radiance or emitted power per unit area per unit solid angle from surface dA_i , $W m^{-2} sr^{-1}$

r = separation distance, m

Table 12.7 Example (hemispherical) solar absorptances and (hemispherical) infrared emittances

Material	Solar absorptance, α_s	Infrared emittance, ϵ_{ir}
Aluminum	0.10–0.15	0.10–0.15
Aluminum (polished)	0.04–0.06	0.04–0.06
Aluminum oxide	0.20–0.30	0.10–0.15
Aluminized Kapton	0.30–0.40	0.55–0.65
Silver	0.05–0.10	0.02–0.05
Silvered Teflon	0.10–0.15	0.70–0.80
Silicon oxide on polished metal	0.05–0.15	0.85–0.95
Titanium	0.75–0.85	0.15–0.25
White paint	0.20–0.25	0.70–0.80
Black paint	0.80–0.90	0.85–0.95
Gold	0.25–0.30	0.02–0.05
Ablative material	0.80–0.90	0.85–0.95
Second surface mirrors	0.05–0.10	0.57–0.80
Solar cells	0.75–0.80	0.80–0.85

β_i, β_j = angles between the normals to the areas and the vector between them

$$d\Omega_k = \frac{\cos \beta_k dA_k}{r^2}, \text{ solid angle steradians, for } k = i \text{ or } j \quad (12.47)$$

The average irradiation or heat flux density incident on area A_j from area A_i follows as

$$\dot{q}_{i \rightarrow j} = E_j = \frac{1}{A_j} \int_{A_i} \int_{A_j} \frac{L_i \cos \beta_i \cos \beta_j}{r^2} dA_i dA_j \quad (12.48)$$

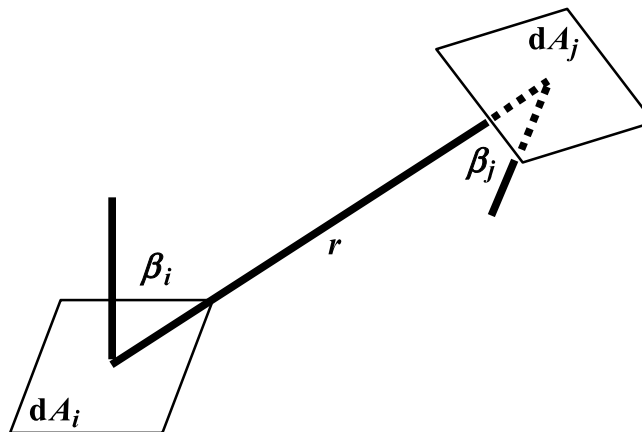


Fig. 12.8 Radiative exchange between two differential areas.

If the L_i is omnidirectional over the hemisphere, it follows from Eq. (3.26) that

$$L_i = \frac{M_i}{\pi} = \frac{\dot{q}_i}{\pi} \quad (12.49)$$

where

$\dot{q}_i = M_i$, heat flux density emanating from A_i

so that

$$\dot{q}_{i \rightarrow j} = E_j = \frac{\dot{q}_i}{A_j} \int_{A_i} \int_{A_j} \frac{\cos \beta_i \cos \beta_j}{\pi r^2} dA_i dA_j \quad (12.50)$$

The total heat flux from A_i incident on A_j is then

$$\begin{aligned} \dot{q}_{i \rightarrow j} A_i &= \frac{\dot{q}_i A_i}{A_j} \int_{A_i} \int_{A_j} \frac{\cos \beta_i \cos \beta_j}{\pi r^2} dA_i dA_j \\ \dot{Q}_{i \rightarrow j} &= \frac{\dot{Q}_i}{A_i} \int_{A_i} \int_{A_j} \frac{\cos \beta_i \cos \beta_j}{\pi r^2} dA_i dA_j \end{aligned} \quad (12.51)$$

where

$\dot{Q}_i = \dot{q}_i A_i$, heat flux emanating from A_i

$\dot{Q}_{i \rightarrow j} = \dot{q}_{i \rightarrow j} A_j$, total heat flux from A_i incident on A_j

This equation defines the *radiative view factor* or *geometric view factor* $F_{i \rightarrow j}$ given by

$$F_{i \rightarrow j} = \frac{\text{Radiant heat flux (irradiance) incident on surface } j \text{ from surface } i}{\text{Total radiant heat flux (radiant exitance) emitted from surface } i} \quad (12.52)$$

so that

$$\dot{Q}_{i \rightarrow j} = \dot{Q}_i F_{i \rightarrow j} \quad (12.53)$$

where

$$F_{i \rightarrow j} = \frac{1}{A_i} \int_{A_i} \int_{A_j} \frac{\cos \beta_i \cos \beta_j}{\pi r^2} dA_i dA_j \quad (12.54)$$

The view factor tells us how much of the radiance exitance or heat that leaves surface i is incident on surface j . From the definition of the view factor, Eq. (12.54), it follows that

$$A_i F_{i \rightarrow j} = A_j F_{j \rightarrow i} \quad (12.55)$$

which is known as the *reciprocity law for view factors*.

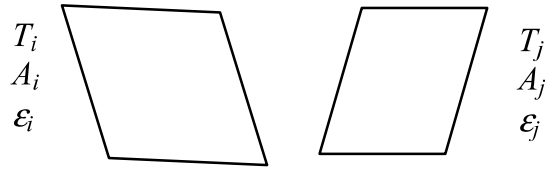


Fig. 12.9 Radiation exchange.

If a surface i is completely enclosed by a set of $j = 1, 2, 3, 4, \dots, n$ surfaces including itself, then all the heat flux leaving surface i must be incident on itself and the other surfaces so that

$$\sum_{j=1}^n F_{i \rightarrow j} = 1 \quad (12.56)$$

The radiation exchange between two surfaces is illustrated in Fig. 12.9. The heat flux that radiates from A_i to A_j and from A_j to A_i , assuming infrared radiation, from the Stefan–Boltzmann equation is

$$\begin{aligned} \dot{Q}_{i \rightarrow j} &= \dot{Q}_i F_{i \rightarrow j} = \varepsilon_i \varepsilon_j \sigma T_i^4 A_i F_{i \rightarrow j} \\ \dot{Q}_{j \rightarrow i} &= \dot{Q}_j F_{j \rightarrow i} = \varepsilon_j \varepsilon_i \sigma T_j^4 A_j F_{j \rightarrow i} \end{aligned} \quad (12.57)$$

The net heat transfer between the two surfaces A_i and A_j , using Eqs. (12.55) and (12.57), is

$$\begin{aligned} \dot{Q}_{i,j} &\equiv \dot{Q}_{i \rightarrow j} - \dot{Q}_{j \rightarrow i} \\ &= \varepsilon_i \varepsilon_j \sigma T_i^4 A_i F_{i \rightarrow j} - \varepsilon_j \varepsilon_i \sigma T_j^4 A_j F_{j \rightarrow i} \\ &= \varepsilon_i \varepsilon_j \sigma T_i^4 A_i F_{i \rightarrow j} - \varepsilon_j \varepsilon_i \sigma T_j^4 A_i F_{i \rightarrow j} \\ &= \varepsilon_i \varepsilon_j \sigma A_i F_{i \rightarrow j} (T_i^4 - T_j^4) \end{aligned} \quad (12.58)$$

12.5.5 Radiative View Factors or Radiative Geometric Factors

The following determination of view factors illustrates how in some cases they can be determined analytically. View factors are tabulated in several references, including Holman (1986). However, in practice it is usually necessary to use computer codes to determine them numerically. Some of the more popular software tools for determining view factors include MATLAB®, TRASYS, VIEW, and MSC NASTRAN Heat Transfer.

12.5.5.1 View factor for a sphere in a sphere. The view factor of a sphere of radius r_i within a concentric sphere of radius r_j illustrated in Fig. 12.10 is determined here. Note that all of the heat flux that leaves the inner sphere will only

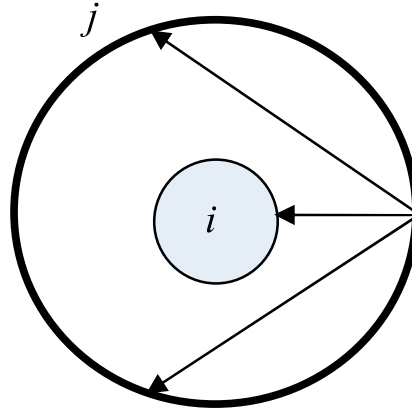


Fig. 12.10 Concentric spheres.

strike the outer sphere, and so the fraction of radiation that strikes the outer sphere is 1, so that by inspection

$$F_{i \rightarrow j} = 1 \quad (12.59)$$

Note that the heat flux from the outer sphere strikes both the inner and outer spheres. Using the reciprocity law, Eq. (12.55), it follows that

$$F_{j \rightarrow i} = F_{i \rightarrow j} \frac{A_i}{A_j} = 1 \frac{4\pi r_i^2}{4\pi r_j^2} = \frac{r_i^2}{r_j^2} \quad (12.60)$$

Since

$$F_{j \rightarrow i} + F_{j \rightarrow j} = 1 \quad (12.61)$$

it follows that

$$F_{j \rightarrow j} = 1 - F_{j \rightarrow i} = 1 - \frac{r_i^2}{r_j^2} \quad (12.62)$$

Consequently, from Eq. (12.53), the heat flux $\dot{Q}_i$ emanating from surface i and incident on surface j , $\dot{Q}_{i \rightarrow j}$, is given by

$$\dot{Q}_{i \rightarrow j} = \dot{Q}_i F_{i \rightarrow j} = \dot{Q}_i \quad (12.63)$$

It follows that

$$\dot{Q}_{j \rightarrow i} = \dot{Q}_j F_{j \rightarrow i} = \dot{Q}_j \frac{r_i^2}{r_j^2} \quad (12.64)$$

so that

$$\dot{Q}_{j \rightarrow j} = \dot{Q}_j F_{j \rightarrow j} = \dot{Q}_j \left(1 - \frac{r_i^2}{r_j^2} \right) \quad (12.65)$$

12.5.5.2 View factor of a half-open channel. The view factors for the surfaces of a long open channel whose external surfaces are insulated and whose cross section is illustrated in Fig. 12.11a are determined. Complete Fig. 12.11a by adding surface A_3 , as in Fig. 12.11b. It follows then that

$$F_{1 \rightarrow 2} + F_{1 \rightarrow 3} = 1 \quad (12.66)$$

From symmetry

$$F_{1 \rightarrow 3} = F_{1 \rightarrow 2} \quad (12.67)$$

so that

$$F_{1 \rightarrow 2} = \frac{1}{2} \quad (12.68)$$

From the reciprocity law, Eq. (12.55), if surface A_1 is of width w and length l , it follows that

$$F_{2 \rightarrow 1} = \frac{A_1}{A_2} F_{1 \rightarrow 2} = \frac{lw}{\frac{1}{4}(\pi w)l} \times \frac{1}{2} = \frac{2}{\pi} \quad (12.69)$$

It follows from Eq. (12.53) that the heat flux exchanges from surface A_1 to surface A_2 , $\dot{Q}_{1 \rightarrow 2}$, and from surface A_2 to surface A_1 , $\dot{Q}_{2 \rightarrow 1}$, in terms of the heat emanating from the surfaces $\dot{Q}_1$, $\dot{Q}_2$, are

$$\dot{Q}_{1 \rightarrow 2} = \dot{Q}_1 F_{1 \rightarrow 2} = \frac{1}{2} \dot{Q}_1 \quad (12.70)$$

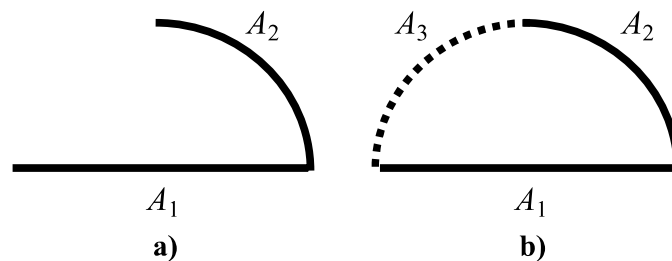


Fig. 12.11 Half-open channel.

$$\dot{Q}_{2 \rightarrow 1} = \dot{Q}_2 F_{2 \rightarrow 1} = \frac{2}{\pi} \dot{Q}_2 \quad (12.71)$$

where

$\dot{Q}_1, \dot{Q}_2$ = total heat flux emanating from surface 1 and 2, respectively

$\dot{Q}_{1 \rightarrow 2}$ = heat flux from surface 1 incident on surface 2

$\dot{Q}_{2 \rightarrow 1}$ = heat flux from surface 2 incident on surface 1

12.5.5.3 View factor of large and small spherical bodies. The view factor between a large spherical body (e.g., Earth) and a small spherical body (e.g., spacecraft) in its vicinity is determined here. The view factor for the transfer of heat flux between the small sphere and the large sphere of radius R_{ls} is determined, and the reciprocity law is then used to determine the view factor between the large sphere and the small sphere separated by distance r . The surface area of the large sphere that intercepts heat flux emanating from the small sphere A_{zone} , as illustrated in Fig. 12.12, is given by

$$A_{zone} = 2\pi Rh = 2\pi R^2 (1 - \cos \eta) \quad (12.72)$$

where

A_{zone} = the area intersected by the large sphere, m^2

r = distance from center of large sphere to small sphere, m

$R = r \cos \eta$

R_{ls} = radius of large sphere, m

η = maximum angle subtended by the sphere

$$\sin \eta = R_{ls} r^{-1} \quad (12.73)$$

The view factor $F_{ss \rightarrow ls}$ from the small sphere to the large sphere is then given by

$$F_{ss \rightarrow ls} = \frac{A_{zone}}{4\pi R^2} = \frac{2\pi R^2 (1 - \cos \eta)}{4\pi R^2} = \frac{1}{2} (1 - \cos \eta) \quad (12.74)$$

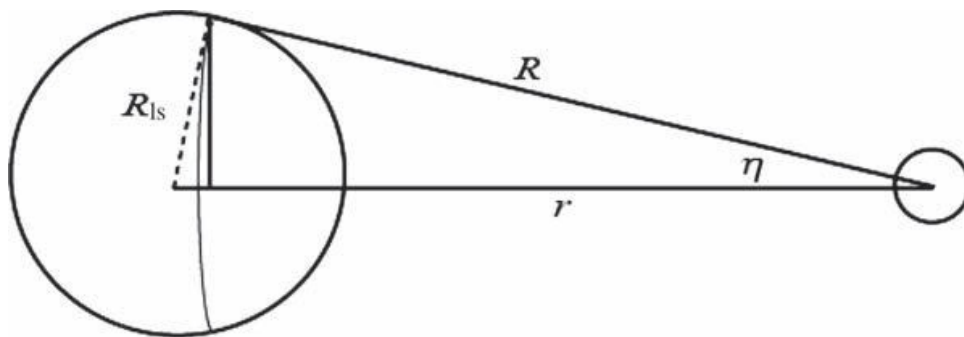


Fig. 12.12 Large and small spherical bodies.

The view factor from the large sphere with surface area A_{ls} to the small sphere with surface area A_{ss} noted by $F_{ls \rightarrow ss}$ is then given by the reciprocity law as

$$A_{ls} F_{ls \rightarrow ss} = A_{ss} F_{ss \rightarrow ls} \quad (12.75)$$

$$F_{ls \rightarrow ss} = \frac{A_{ss}}{A_{ls}} F_{ss \rightarrow ls} = \frac{1}{2} \frac{A_{ss}}{A_{ls}} (1 - \cos \eta) \quad (12.76)$$

where

A_{ss} = surface area of small sphere, m^2

A_{ls} = surface area of large sphere, m^2

The total radiation incident on the small sphere $\dot{Q}_{ls \rightarrow ss}$ in terms of the heat flux per unit area emitted by the large sphere $\dot{q}_{ls}$ is

$$\dot{Q}_{ls \rightarrow ss} = F_{ls \rightarrow ss} \dot{q}_{ls} A_{ls} = \frac{1}{2} (1 - \cos \eta) \dot{q}_{ls} A_{ss} \quad (12.77)$$

where

$\dot{q}_{ls}$ = heat flux per unit area emitted by large sphere, $W\ m^{-2}$

$\dot{Q}_{ls \rightarrow ss}$ = heat flux radiated from large to small sphere, W

12.5.5.4 View factor of large and small spherical bodies at long distance. The view factor between a large spherical body and a small spherical body at a large separation distance is determined here. It follows directly from Eqs. (12.76) and (12.73) that

$$\begin{aligned} F_{ls \rightarrow ss} &= \frac{1}{2} \frac{A_{ss}}{A_{ls}} \left(1 - [1 - (R_{ls}/r)^2]^{1/2} \right) \\ &\approx \frac{1}{4} \frac{A_{ss}}{A_{ls}} (R_{ls}/r)^2 \end{aligned} \quad (12.78)$$

where

A_{ss} = surface area of small sphere, m^2

A_{ls} = surface area of large sphere, m^2

r = distance from center of large sphere to center of small sphere, m

R_{ls} = radius of large sphere, m

It follows from Eq. (12.77) that the heat flux incident on the small sphere from the large sphere is

$$\begin{aligned} \dot{Q}_{ls \rightarrow ss} &= F_{ls \rightarrow ss} \dot{q}_{ls} A_{ls} \\ &= \frac{1}{4} \frac{A_{ss}}{A_{ls}} \left(\frac{R_{ls}^2 \dot{q}_{ls}}{r^2} \right) A_{ls} = \left(\frac{R_{ls}}{r} \right)^2 \dot{q}_{ls} A_{p,ss} = \dot{q}_{ls} A_{p,ss} \sin^2 \eta \end{aligned} \quad (12.79)$$

where

$A_{p,ss}$ = planar area of spherical satellite, m^2
 $\dot{q}_{ls}$ = heat flux density at surface of large sphere, $W\ m^{-2}$

$$\sin \eta = \frac{R_{ls}}{r} \quad (12.80)$$

12.5.5.5 View factor between diffuse large spherical body and small flat plate. The view factor between a large diffusely radiating spherical body (e.g., Earth or sun) and one side of a small flat plate, illustrated in Fig. 12.13, whose normal is inclined at an angle λ from the direction to the center of the sphere is given here. The view factor from the sphere to the plate $F_{s \rightarrow p}$ and the heat flux incident on the plate $\dot{Q}_{s \rightarrow p}$ in terms of the heat radiated by the sphere $\dot{q}_s$ is by inspection

$$F_{s \rightarrow p} = \frac{A_p \cos \lambda}{4\pi r^2} = \frac{A_p \cos \lambda}{A_s} \left(\frac{R_s}{r} \right)^2 = \frac{A_p}{A_s} \cos \lambda \sin^2 \eta \quad (12.81)$$

$$\dot{Q}_{s \rightarrow p} = F_{s \rightarrow p} \dot{q}_s A_s = \dot{q}_s A_p \cos \lambda \sin^2 \eta \quad (12.82)$$

where

A_p = area of plate, m^2
 A_s = surface area of sphere on m^2
 $\dot{q}_s$ = heat flux density of sphere, $W\ m^{-2}$
 r = distance between center of sphere and plate, m
 R_s = radius of sphere, m
 λ = angle between normal to the plate and direction of center of sphere
 η = maximum angle subtended by the sphere from the plate
 $\sin \eta = R_s/r$

12.5.5.6 Power incident on spherical body from Earth albedo. The power incident on a spherical body from the Earth, $\dot{Q}_{a \rightarrow s}$, can be expressed by

$$\dot{Q}_{a \rightarrow s} = A_b S_e F_{a \rightarrow s}(h, \theta_s) A_s = 4 A_b S_e F_{a \rightarrow s}(h, \theta_s) A_p \quad (12.83)$$

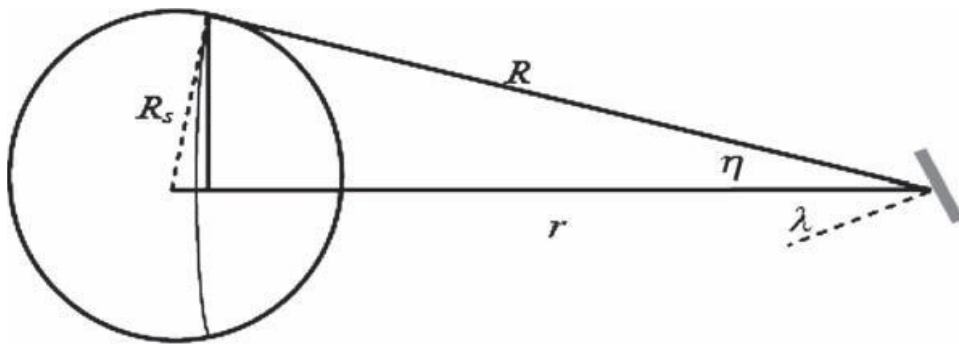


Fig. 12.13 Geometry of sphere and flat plate.

where

- A_b = Bond albedo of sphere
- A_s = surface area of sphere, m^2
- A_p = cross-sectional area of sphere, m^2
- h = altitude, m
- $F_{a \rightarrow s}$ = view factor from Bond albedo surface to sphere
- $\dot{Q}_{a \rightarrow s}$ = heat flux from Bond albedo surface to sphere
- S_e = solar constant at Earth distance, W m^{-2}
- θ_s = angle between the vectors to the sun and the spacecraft from the center of the reflecting body

The ordinate of Fig. 12.14, from NASA TN D-1099 (Cunningham 1961), gives the heat incident on the sphere per cross-sectional area,

$$\left(\frac{\dot{Q}_{a \rightarrow s}(h, \theta_s)}{A_p} \right)$$

so the view factor can be obtained from the figure using Eq. (12.83):

$$F_{a \rightarrow s}(h, \theta_s) = \frac{1}{4 A_b S_e} \left(\frac{\dot{Q}_{a \rightarrow s}(h, \theta_s)}{A_p} \right) \quad (12.84)$$

To generate Fig. 12.14, the Bond albedo used was a mean value of 0.34 and the solar constant used was 1343 W m^{-2} . Consequently, the incident power in Fig. 12.14

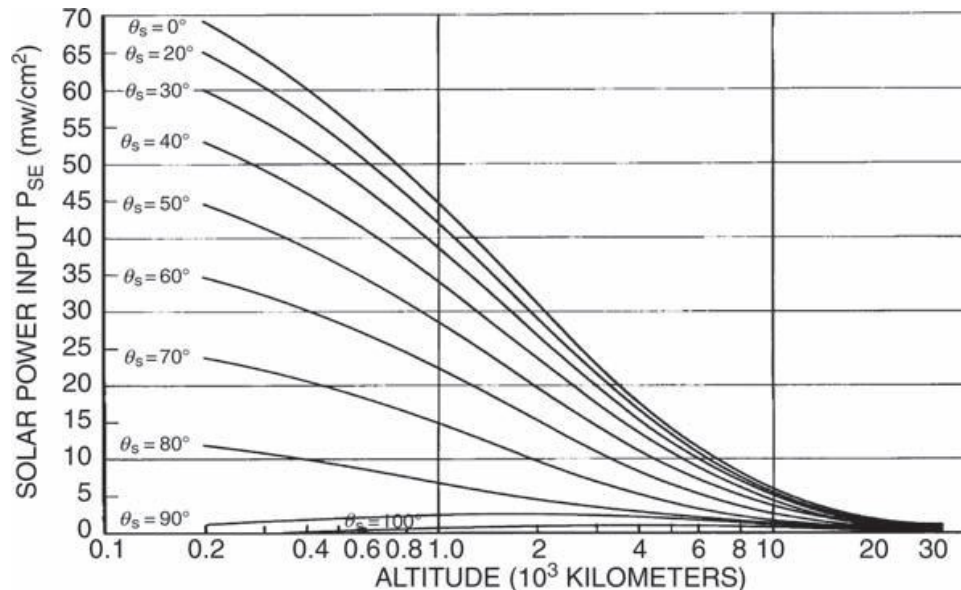


Fig. 12.14 Heat incident from Earth albedo on a spherical body per cross-sectional area as a function of solar orientation. [From Cunningham (1961); courtesy of NASA.]

can be scaled by the Bond albedo and solar constant as

$$\dot{Q}_{\text{Albedo heat from Earth}} = A_p \left(\frac{\dot{Q}_{a \rightarrow s}(h, \theta_s)}{A_p} \right) \left(\frac{A_b}{0.34} \right) \left(\frac{S_e}{1343} \right) \quad (12.85)$$

where $\left(\frac{\dot{Q}_{a \rightarrow s}(h, \theta_s)}{A_p} \right)$ is the ordinate in Fig. 12.14.

12.5.5.7 Earth reflected solar radiation incident on a rotating flat plate. The geometry of a rotating flat plate in space exposed to Earth albedo is illustrated in Fig. 12.15, and the average incident radiation from the Bond albedo $\dot{Q}_{a \rightarrow p}$ is given by Fig. 12.16. In this case, the axis of rotation of the plate is pointing toward the center of the Earth, where the angle of the sun with respect to the rotation axis of the plate is θ_s , and the angle between the normal to the plate and the rotation axis is λ . When $\lambda = 0$ deg, the normal to the plate is pointing at the center of the Earth. The heat incident on the flat plate can be represented by

$$\dot{Q}_{a \rightarrow p} = A_b S_e F_{a \rightarrow p}(h, \lambda, \theta_s) A_p \quad (12.86)$$

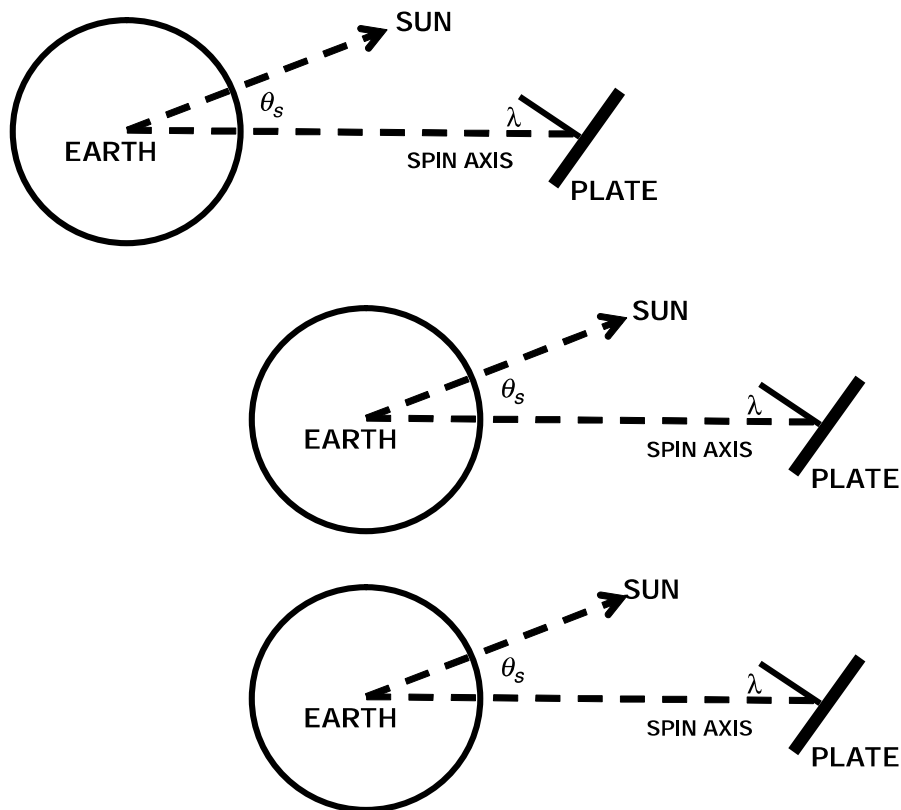


Fig. 12.15 Geometry of a rotating flat plate exposed to Earth Bond albedo.

where

A_b = Bond albedo of plate

A_p = area of plate, m^2

h = altitude, m

S_e = solar constant, W m^{-2}

$F_{a \rightarrow p}$ = view factor from Earth Bond albedo illumination to flat plate

$\dot{Q}_{a \rightarrow p}$ = heat flux from Earth Bond albedo illumination to flat plate

λ = angle between the normal to the plate and the direction toward the center of the Earth, deg

θ_s = angle between the vectors to the sun and the spacecraft from the center of Earth

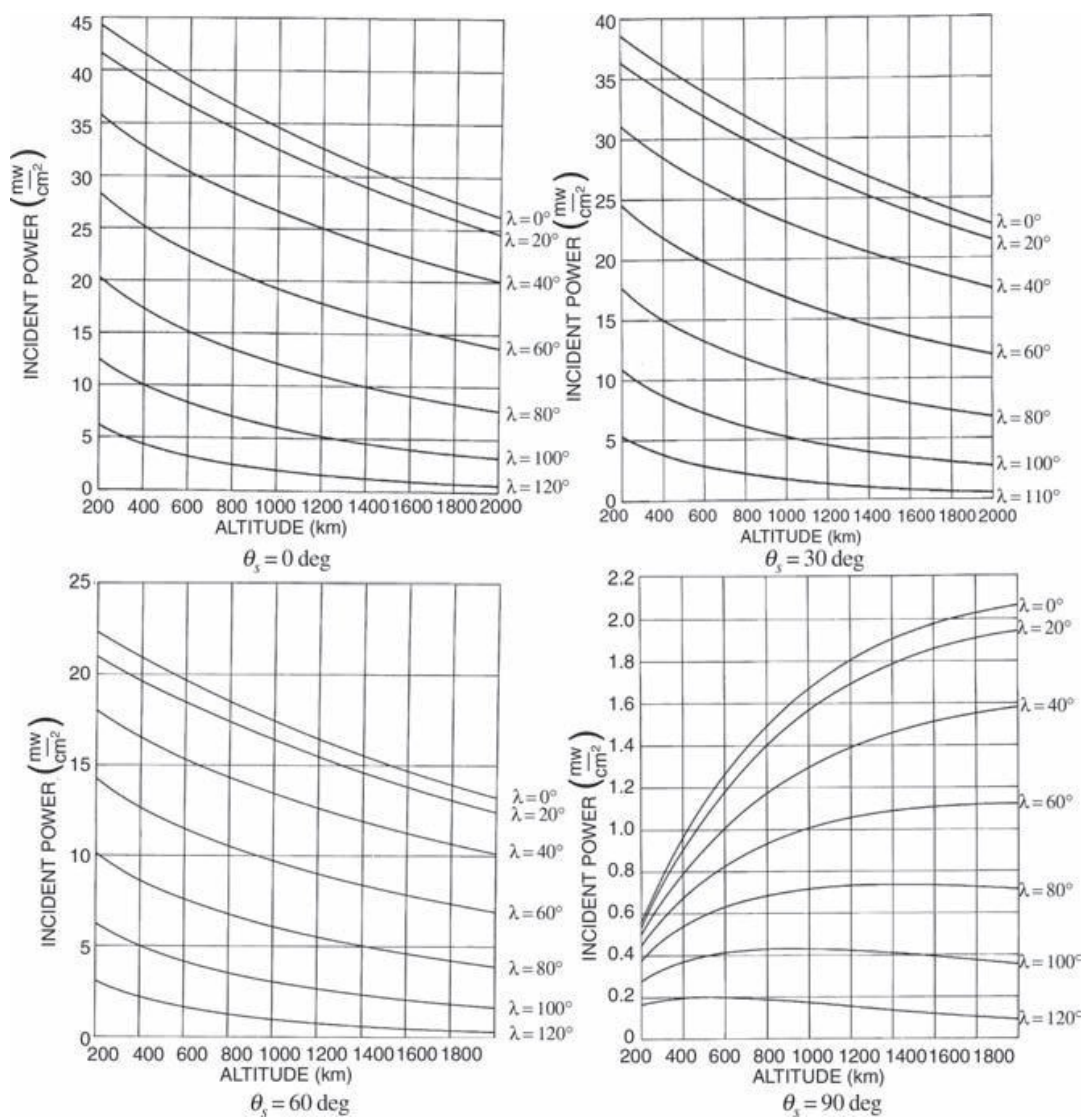


Fig. 12.16 Incident power for spinning flat plate inclined by angle λ . [From Cunningham (1963); courtesy of NASA.]

The ordinate of Fig. 12.16 gives the heat per unit area incident on the plate area,

$$\left(\frac{\dot{Q}_{a \rightarrow p}(h, \lambda, \theta_s)}{A_p} \right)$$

so the view factor can be determined as

$$F_{a \rightarrow p}(h, \theta_s) = \frac{1}{A_b S_e} \left(\frac{\dot{Q}_{a \rightarrow p}(h, \lambda, \theta_s)}{A_p} \right) \quad (12.87)$$

To generate Fig. 12.16, the Bond albedo used was a mean value of 0.34, and the solar constant used was a mean value of 1395 W m^{-2} . Consequently, the incident power in Fig. 12.16 can be scaled by the bond albedo and solar constant as

$$\dot{Q}_{\text{Albedo heat from Earth}} = A_p \left(\frac{\dot{Q}_{a \rightarrow s}(h, \lambda, \theta_s)}{A_p} \right) \left(\frac{A_b}{0.34} \right) \left(\frac{S_e}{1395} \right) \quad (12.88)$$

where

$$\left(\frac{\dot{Q}_{a \rightarrow s}(h, \lambda, \theta_s)}{A_p} \right)$$

is incident power from the ordinate of Fig. 12.15.

12.6 Spacecraft Thermal Control Components

12.6.1 Thermal Control Coatings

The simplest technique to modify a surface's thermal characteristics is to cover it with paint or a layer of more suitable material. Spacecraft designers can make use of many different kinds of coatings, ranging from relatively simple-to-apply paints to more sophisticated chemically or physically produced coatings. Surface roughness and thickness of a coating will have a significant effect on its infrared emittance and solar absorptance. When the wavelength of the irradiance is comparable to the dimensions of the surface roughness or imperfections, multiple scattering can occur that will affect the solar absorptance and infrared emittance. The primary disadvantages of coatings are their susceptibilities to degradation of solar absorptance, infrared emittance, ability to adhere to the surfaces, and electrical conductivity. These changes can occur while in orbit due to deposition of outgassed materials, electromagnetic photons, trapped radiation, and atomic oxygen, and before launch due to ground handling. Coatings should be grounded to eliminate electrostatic discharge when placed on nonconductive substrates by the introduction of a thin conductive

film between the coating and the substrate. The factors taken into account in selecting a coating include optical properties and their uncertainties, ease of application, effect of handling, effects of ground environment, outgassing, toxicity, effect of the space environment including particle radiation and ultraviolet photons, mechanical properties, and cost. New coatings should be tested on the specific substrate on which they will be placed and then exposed to conditions simulating the mission.

12.6.2 Second Surface Mirrors

A reflective coating can be deposited on either the first or second side of a transparent surface forming, respectively, a first surface mirror or a second surface mirror. Consequently, a first surface mirror reflects off the front side of a mirror while a second surface mirror reflects off the back side. In a second surface mirror, solar radiation travels through the transparent surface and reflects off the mirrored second surface, and so it is a good reflector of solar radiation. At the same time, infrared heat conducts through the reflective second surface into the transparent layer. If that layer is selected to be a good radiator of radiation, the second surface mirror will provide low solar absorptance and high infrared emittance, as illustrated in Fig. 12.17. Second surface mirrors are often used for the radiating surface of a radiator or louver. A typical second surface mirror is made of a transparent layer of Teflon with the second surface silvered.

12.6.3 Multilayer Insulation

Multilayer insulation (MLI) uses multiple heat radiation barriers to provide high-performance insulation. Each individual barrier consists of a thin polymer film that reflects a significant amount of heat flux. An example of the construction of a MLI blanket is illustrated in Fig. 12.18.

The materials used in MLI blankets are described in Table 12.8 and include Kapton, Mylar, Tedlar, Teflon, and Beta cloth with Dacron or Nomex separation layers.

Kapton, Mylar, and Teflon adhesive tapes are used to attach blankets of MLI together or to the structure and are stitched at intervals with quartz or glass thread.

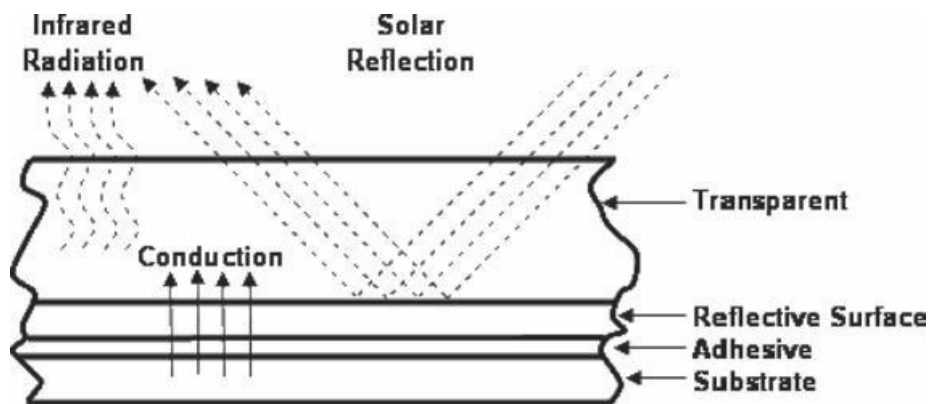


Fig. 12.17 Second surface mirror.

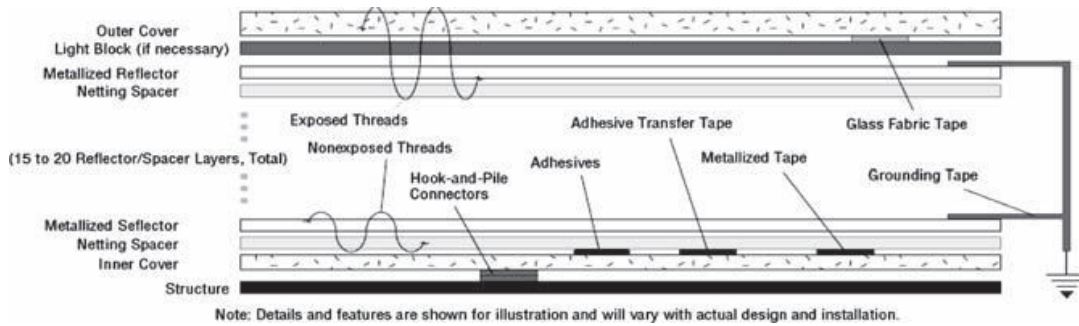


Fig. 12.18 Components of a MLI blanket. [From Finckenor and Dooling (1999); courtesy of NASA.]

Table 12.8 Materials used in MLI

Material	Characteristics
Kapton	Polyimide film Stable over wide temperature ranges Chemical and radiation resistant Electrical insulator Susceptible to wear
Mylar or Melinex	Polyethylene terephthalate (PET) Strong polyester film High tensile strength Resistant to chemical and weather Electrical insulator
Tedlar	Polyvinyl fluoride (PVF) film Has tough, cleanable surface Resistant to weather and chemicals Resists fading by ultraviolet radiation Electrical insulator
Teflon	Polytetrafluoroethylene (PTFE) High chemical resistance Low friction Electrical and thermal insulator
Beta cloth	Fiberglass cloth impregnated with PTFE Teflon Used for outer MLI blankets Tear resistant Resistant to atomic oxygen Can be aluminum coated
Dacron netting	Used for separation between layers Polyester material Temperature range -70 to $+120^{\circ}\text{C}$
Nomex netting	Used for separation between layers Nomex aramid material Temperature range -70 to $+120^{\circ}\text{C}$

Source: From NASA TP-1999-209263 (Finckenor and Dooling 1999).

Metalized tape with aluminum or gold can be used to electrically ground the MLI. Organic hook and pile fasteners made of Velcro, Nomex, nylon, and polyester are used on surfaces not generally exposed to the environment with stainless steel hook and pile fasteners used on exposed surfaces. Hand ties are also used to attach the MLI blanket to the structure.

The heat transfer through multilayer insulation can be expressed in terms of an equivalent emittance ϵ_{eq} and the temperature difference between the inside and outside, T_i^4 , T_o^4 , as

$$\dot{q} = \epsilon_{eq} \sigma |T_i^4 - T_o^4| \quad (12.89)$$

where

$$\sigma = \text{Stefan-Boltzmann constant, } \sigma \equiv 2\pi^5 k^4 / 15 h^3 c^2, \text{ W m}^{-2} \text{ K}^{-4}$$

The equivalent emittance between two infinite parallel surfaces with different emittances and temperatures, as illustrated in Fig. 12.19a, is given by Karam (1998) as

$$\epsilon_{eq} = (\epsilon_1^{-1} + \epsilon_2^{-1})^{-1} \quad (12.90)$$

The equivalent emittance of three infinite parallel surfaces with different emittances, as illustrated in Fig. 12.19b, is given by Karam (1998) as

$$\epsilon_{eq} = [\epsilon_1^{-1} + \epsilon_2^{-1} + (1 - \epsilon_{s1}) \epsilon_{s1}^{-1} + (1 - \epsilon_{s2}) \epsilon_{s2}^{-1}]^{-1} \quad (12.91)$$

The equivalent emittance of n infinite parallel surfaces with the same emittances, as illustrated in Fig. 12.19c, is given by Karam (1998) as

$$\epsilon_{eq} = [(n-1)(2\epsilon^{-1} - 1)]^{-1} \quad (12.92)$$

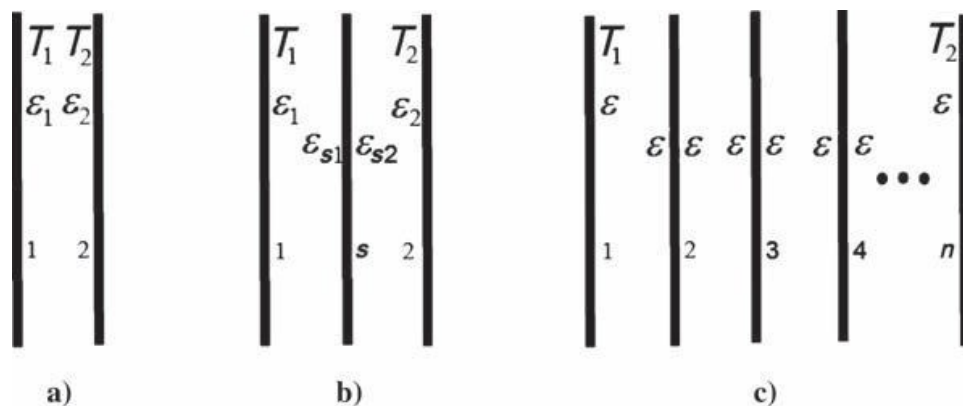


Fig. 12.19 Multilayer blanket geometry.

For values of ε_1 and ε_2 in the range of 0.03–0.05, the effective emittance of a thermal blanket of 20 layers of MLI ranges between 0.0008 and 0.0013.

For effective performance, the residual pressure in the layers of the MLI must be less than 10^{-4} mm Hg. To achieve this and to prevent damage to the thermal blanket during ascent, it is necessary to provide adequate venting. This is generally accomplished by not sealing the edges of the blanket and providing venting holes throughout the layers of the MLI.

On the sun-facing side of a spacecraft, a gold-colored MLI thermal blanket is often used. The outer surface is a thin film of Kapton with an inner surface that is silver coated with the combination giving it its gold color. The silvered side is grounded to reduce electric charge buildup. This configuration acts as a second surface mirror with low solar absorptance and high infrared emittance. Thus it is good at rejecting solar radiation and emitting spacecraft infrared thermal radiation.

On the space-facing side of a spacecraft, a black-colored MLI blanket is often used. The outer surface is a thin film of Kapton impregnated with carbon that gives it its black color, and because carbon is conductive, it can be grounded to reduce electric charge buildup. This configuration provides high solar absorptance and low thermal emittance that reduces the net loss of heat from the spacecraft.

12.6.4 Radiators and Louvers

The most common and simplest way to dissipate internal heat flux is through a radiator that is generally an aluminum surface that is part of the structure exposed to the environment. Often its surface is finished with a coating that provides high infrared emittance >0.9 and low solar absorptance <0.3 . If more heat flux dissipation is required, a fin type radiator can be employed where multiple surfaces are exposed. Deployable radiators can provide more exposed area than can normally fit within the launch fairing, so that even more heat can be dissipated. For example, the International Space Station uses a deployable radiator system that uses pumped liquid ammonia as the refrigerant. The radiators consist of 14 panels with each panel measuring 1.8×3 m for a total radiator area of 156 m^2 .

Louvers are devices where the exposed area is changed by either a passive or active mechanism in response to the magnitude of a parameter, typically temperature. A typical passive louver, as illustrated in Fig. 12.20, uses a series of Venetian blind shutters with each controlled by a bimetallic coil. The radiator can be blocked below a certain temperature and adjust itself according to the temperature.

An active louver uses electrical power to control the heat flux radiated. One type of active louver is a radiator with a flexible surface over it that can be retracted, like a shade on a roller by an electric motor, according to temperature.

12.6.5 Phase-Change Materials

Phase-change materials store heat by changing phase, through latent heat of sublimation or fusion. They are generally useful when there is significant energy dissipated by a subsystem for a brief interval. After storing heat, they can then release it over a longer interval of time, limiting the maximum temperature. They

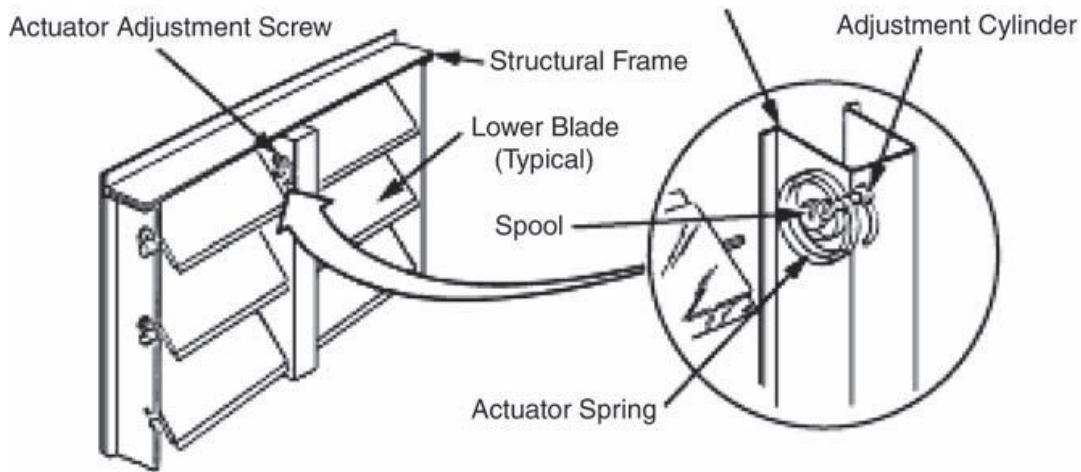


Fig. 12.20 Orbital Sciences passive louver assembly. [Courtesy of Orbital Sciences Corporation (2007).]

Table 12.9 Phase-change material properties

Material	Transition	Transition temperature, °C	Latent heat, kJ kg ⁻¹	Density, kg m ⁻³
Paraffins				
Tetradecane	Solid–liquid	5.5	0.228	771
Hexadecane	Solid–liquid	16.7	0.237	774
Octadecane	Solid–liquid	28.0	0.244	774
Nonparaffin organics				
Polyethylene glycol 600	Solid–liquid	68 to 77	146	1100
Acetic acid	Solid–liquid	16.7	187	1050
Tristearin	Solid–liquid	56	190.8	862
Salt hydrates				
Sodium hydrogen phosphate dodecahydrate	Solid–liquid	36	280	1520
Lithium nitrate trihydrate	Solid–liquid	29.88	296	1550
Barium hydroxide octahydrate	Solid–liquid	78	301	1720
Metals				
Cerrobend eutectic	Solid–liquid	70	32.6	9400
Gallium	Solid–liquid	30.0	80.3	5903
Cerrolow eutectic	Solid–liquid	58	90.9	8800
Solid–solid				
Diaminopentaerythritol	Solid–solid	68	184	—
2-amino-2methyl-1,3-propanediol	Solid–solid	78	264	—
Pentacrythritol	Solid–solid	184	301	1350
Others				
Water	Solid–liquid	0.0	333.4	999.8
Aluminum chloride	Solid–liquid	192.4	272	2440
Dowtherm A	Solid–liquid	12	97.9	1060

Source: From NASA CR-61363 (Hale et al. 1971).

are limited in their capacity, and once their phase changes, they cannot store additional heat. Materials that rely on the latent heat of sublimation are generally open loop where the gas is released. Materials that rely on the latent heat of fusion are generally reusable. A simple phase-change device is a wax-type substance in a container connected to the heat generating device by a conduction plate. An ideal phase-change material would have a high heat of fusion, high thermal conductivity, high specific heat density, long-term reliability during repeated cycles, low volume change during phase transition, and reversible phase change. Properties of potential phase-change materials are given in Table 12.9.

12.6.6 Electrical Resistance Heaters

A common approach to a thermal control system is to bias the system cold and use thermostatically controlled electric resistance heaters to maintain the required operating temperature. While a straightforward approach, this can be costly in power. However, most spacecraft systems do utilize electrical resistance heaters.

12.6.7 Mechanical Elements, Thermal Doublers

The mechanical structure of the instrument or spacecraft has a significant effect on the conduction and radiation paths for heat transfer within the instrument or spacecraft. Base plates and fixation elements can be used to conduct heat away from its source. Thermal doublers are mechanical heat sinks made of highly thermal conductive materials used to provide conductive pathways to transport heat for dispersal by other means, such as radiators or louvers. Unfortunately, they add mass to the system. Braids of highly conductive material, such as copper, can also be used to transport heat flux away from its source. Their mechanical flexibility is an advantage during integration.

12.6.8 Heat Pipes

Heat pipes are closed systems that provide highly effective conductive pathways, either passively or actively, in which heat flux is transported by evaporation and condensation. A heat pipe consists of a working fluid, a wick or capillary structure, and a container made up of three sections: an evaporator section, an adiabatic section, and a condenser section, as illustrated in Fig. 12.21. The wick is

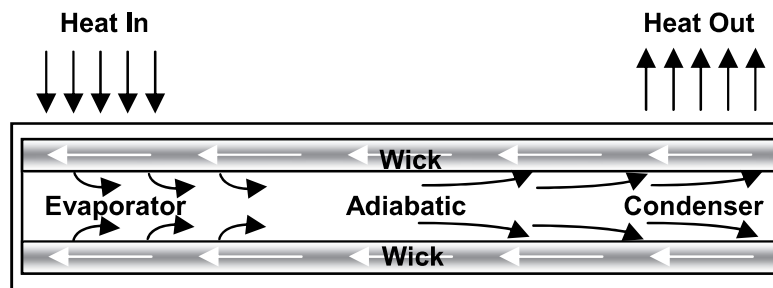


Fig. 12.21 Schematic of a heat pipe.

generally on the inner surface of the container except at the ends of the evaporator and condenser ends. In the evaporator section, heat is absorbed by a liquid to form a vapor in which the amount absorbed is a function of the latent heat of vaporization of the working fluid. The increase in vapor pressure forces the vapor into the adiabatic section. The adiabatic section transports the vapor from the evaporator section to the condenser section with minimal drop in pressure and negligible heat loss. In the condenser section, the vapor is cooled and condensed back to a liquid, giving up its heat of vaporization. The liquid is then returned to the condenser section by a wick that employs porous capillaries.

A passive isothermalizer heat pipe is capable of transporting heat in either direction, dependent on the direction of the temperature difference. By introducing passive or active devices, the heat pipe can be made to operate in only one direction or to vary the amount of heat, the latter a *variable conductance heat pipe*. A common design of a variable conductance heat pipe is to insert a bellows containing a nonconducting gas under pressure at the condenser end. As the temperature of the conducting gas increases, the increased vapor pressure compresses the bellows, exposing more of the condenser. The opposite occurs as the conducting vapor temperature and pressure decrease. When precise control is required, a variable conductance heat pipe can be used with an integrated electronic control. The container material is selected based on its strength-to-mass ratio, thermal conductivity, and workability. A high thermal conductivity ensures a small temperature drop between the evaporator section and the heat source and the condenser section and the region to which heat is rejected. Workability includes machinability, weldability, and ductility for ease of fabrication.

The working fluid should have a high latent heat of vaporization, high thermal conductivity, low viscosity, be wettable, and have high surface tension. The larger the latent heat of vaporization, the greater the heat transfer rate for a given sized heat pipe. A high thermal conductivity reduces the potential of large radial temperature gradients, reducing the potential for vaporization in the wick. Fluid flow is enhanced if the vapor and liquid viscosities are low. The greater the wettability, the greater the potential for capillary action in the wick giving an increased liquid

Table 12.10 Sample heat pipe working fluids

Fluid	Boiling temperature, K	Latent heat of vaporization, kJ kg^{-1}	Specific heat, $\text{kJ kg}^{-1} \text{K}^{-1}$	Equivalent temperature change, latent heat/specific heat, K
Acetone	329	518	2.15	241
Ammonia	240	1180	4.80	246
Lithium	1615	19330	4.27	4526
Methane	111	509	3.45	147
Oxygen	90	213	1.90	112
Potassium	1032	1920	0.81	2370
Sodium	1152	3600	1.38	2608
Water	373	2260	4.18	541

Source: From NASA (undated).

flow rate. Higher values of surface tension result in a greater capillary flow rate. The wick or capillary structure is made of metals such as aluminum, copper, and nickel; fibrous material such as ceramic and carbon fibers; and other materials such as felts. Various pore sizes measured in microns are employed. The smaller the pore size, the greater the pumping pressure, but the greater the flow resistance, so that it is necessary to optimize pore size for the particular working fluid employed. The capacity of a heat pipe is generally limited by the velocity by which the fluid can flow through the wick. Sample working fluids and their properties are provided in Table 12.10.

A loop heat pipe is a passive two-phase (gas–liquid) thermal transfer device that uses capillary forces to circulate the working media. They have characteristically demonstrated the transport of significantly more heat than a conventional heat pipe. Figure 12.22 illustrates the schematic of a loop heat pipe.

12.6.9 Radioisotope Heater Units

When it is not feasible to utilize electrical resistance heaters in a thermal control system, an alternative is the radioisotope heater unit (RHU). As an example, the Cassini-Huygens spacecraft that orbited Saturn used 82 RHUs and the Huygens probe it carried used 35 for thermal control purposes.

The currently available RHU, as illustrated in Fig. 12.23, is relatively compact, measuring 3.2 cm in length and 2.6 cm in diameter. Its mass is 40 g including the fuel pellet, which is about the size and shape of a pencil eraser and has a mass of about 2.7 g. Heat is generated by the decay of plutonium-238, which has a half-life of 87.7 years. The RHUs neither have moving parts nor require power, and so they are extremely reliable, their heat output is predictable, and they are resistant to radiation and meteoroid damage. They do provide a radiation environment that may cause problems with ultrasensitive nearby electronics and operationally pose access issues before launch once they are installed. In case of an accident, containment of the radioactive material is by the multiple-layer metal containers that are resistant to heat and stress.

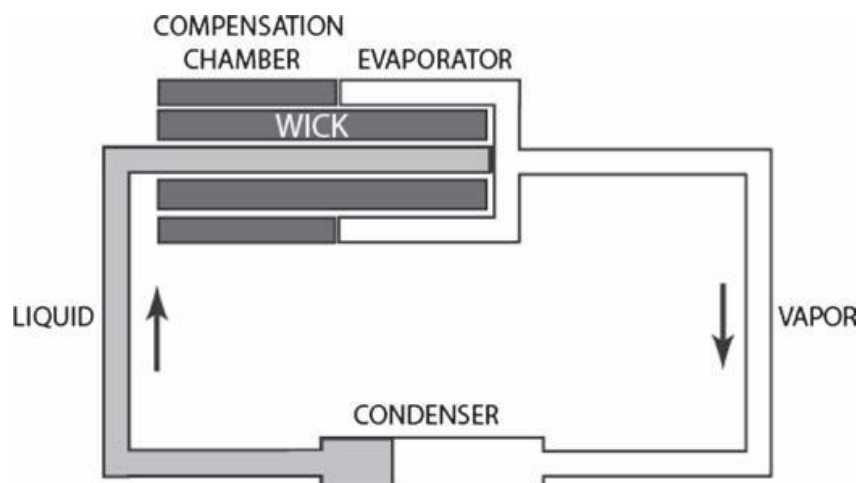


Fig. 12.22 Loop heat pipe schematic.

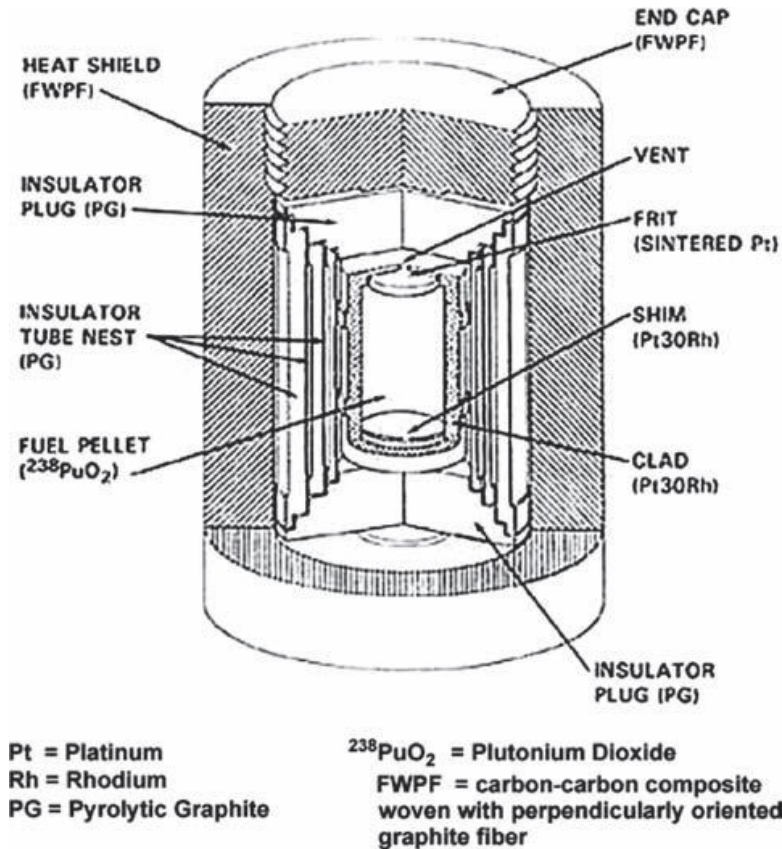


Fig. 12.23 Radioisotope heater units. [Courtesy of the U.S. Department of Energy (2007).]

12.6.10 Pumped-Loop System

A pumped-loop system utilizes a fluid to transport heat from its source to a space radiator. Working fluids used in past missions include ammonia on the International Space Station, CFC-11 (chlorofluorocarbon) on the Mars Exploration Rover, water on the shuttle, and methanol and water on Skylab. The working fluid is pumped, generally by a centrifugal pump, under thermostatic control.

12.7 Spacecraft Design

The thermal design of a spacecraft or a space instrument is an iterative process carried out in parallel with the development of the other subsystems. The specifications to be satisfied are the temperature limits for the various subsystems and components as discussed in Section 12.1.

The development process is outlined in Fig. 12.24. Critical in the design is the internal power dissipation, placement of the various subsystems, design of the structure, thermal analyses, fidelity of the thermal model, and the testing carried out to validate the design and the thermal model. The structural design is important as heat flux is disturbed between the interior and the outer surfaces of the instrument or spacecraft by radiation and conduction through the structural members. Subsystem placement is important to distribute the internal heat flux dissipation to advantage.

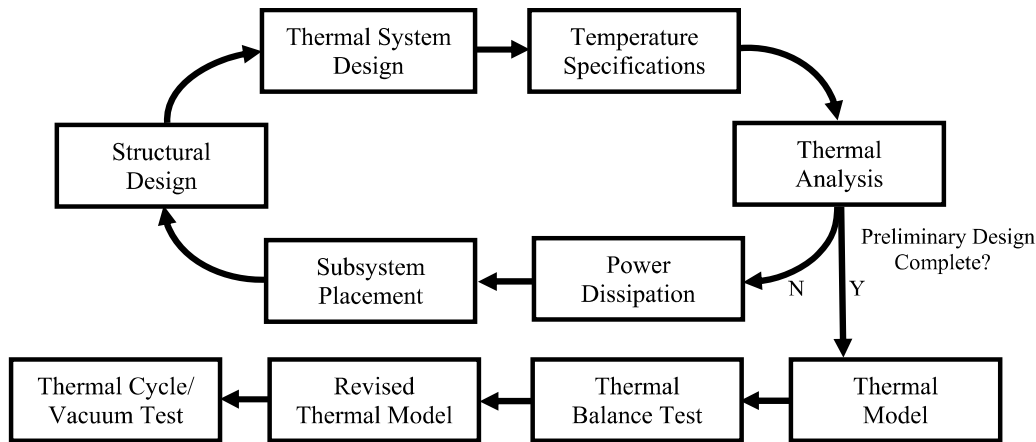


Fig. 12.24 Thermal system development process.

The selection of the components of the thermal control system includes the many options discussed in Section 12.6. Thermal control systems generally consist of both passive and active components. Passive components utilize no power, are inherently more reliable, generally are less costly, and can function when spacecraft power is limited or not available. Candidate passive and active components are identified in Table 12.11. Typically, the thermal control system is designed with passive components, and if the requirements cannot be satisfied, active components are added. This is often accomplished by biasing the spacecraft or instrument to be colder than desired and using resistance heaters to attain the operating temperatures and adding other selective active components as required.

In carrying out the design of a thermal control system, if during eclipse the minimum temperature is too low then the options are to decrease the infrared emittance or add resistance heaters. Increasing the solar absorptance is not an option, as the sun does not provide heat during eclipse. During full sunlight, if the maximum temperature is too high, the options are to increase the infrared emittance, decrease the solar absorptance, add radiators, and reduce the exposed area by adding a sun shield or shade.

Spacecraft and spacecraft component environmental test requirements are specified in MIL-STD-1540D (U.S. Department of Defense 1999) and NASA-STD-7002

Table 12.11 Thermal control systems components

Passive	Active
Thermal insulation	Resistance heaters
Surface coatings (finishes, paint, second-surface mirrors)	Variable-conductance heat pipes
Thermal shields	Heat pumps and pumped fluid loops
Thermal storage devices (e.g., phase-change devices, cryogenes)	Thermoelectric or thermionic cooling
Constant or variable conductance heat pipes	Active radiators or louvers
Structural doublers	
Radiators	

(NASA 1996). For spacecraft with no thermal control or only passive thermal control, the recommended minimal thermal uncertainty margin is 17°C prior to achieving a validated thermal model, which can be reduced to 11°C after the analytical model is validated by a thermal balance test. For spacecraft with active thermal control, a heater load margin of 25% may be substituted for the thermal control margin. Thermal testing generally includes a thermal balance vacuum test followed by a thermal cycle and/or thermal vacuum test. The objectives of the thermal balance vacuum test include 1) verifying the capability of the thermal control system, 2) verifying the thermal model, 3) assessing the effectiveness of the thermal control system components, 4) measuring thermal deformations if important, and 5) updating flight temperature predictions. The objectives of the thermal cycle and/or thermal vacuum test include the demonstration of the ability to 1) verify the thermal specifications in the operating environments, 2) verify the thermal design margins in the operating environment, and 3) withstand the thermal stressing from temperature changes.

Thermal cycle and *thermal vacuum tests* differ in that the latter is carried out in a vacuum while the former is carried out in air. Thermal cycle testing is used to demonstrate the ability of the system or spacecraft to tolerate the consequences of thermal cycling during operations. A thermal cycle test is easier to carry out than a thermal vacuum test, and a combination of a thermal cycle test with a reduced thermal vacuum test may be selected in place of a thermal vacuum test.

The number of cycles in thermal cycle tests and thermal vacuum tests or combinations and the temperature ranges for acceptance and qualification tests are given in Table 12.12. A *qualification test* is carried out to assess design margins and the test article is not flown. An *acceptance test* is carried out on flight hardware to ensure it meets the design specifications and is typically performed at lower levels than a qualification test. A *thermal balance test* is used to verify the adequacy of the thermal design and the capability of the thermal control system. The results of the thermal balance test are often used to establish the temperature goals of the thermal vacuum test. Often the first cycles of the thermal vacuum test are used as the thermal balance tests. During a cycle, the ambient temperature is raised to the specified high temperature and stabilized for a high-temperature soak. Following this, the temperature is reduced to the specified minimum temperature for a low-temperature soak. The temperature is then returned to ambient

Table 12.12 Temperature and cycles for thermal cycle (TC) and thermal vacuum (TV) tests

Test	Temperature range	Number of cycles					
		TC and TV		TV only		TC only	
		Required	Maximum	Required	Maximum	Required	Maximum
Acceptance test	105°C	8.5 and 4	17 and 8	1	2	12.5	25
Qualification test	125°C	53.5 and 25		6		78.5	

Source: From MIL-STD-1540D (U.S. Department of Defense 1999).

temperature to complete one cycle. A thermal soak at the extremely high or low temperature conditions is generally for a duration of 6 h. Thermistors are used to assess the temperature of critical components during tests and are telemetered to the control station when the spacecraft is operational.

Thermal vacuum tests generally consist of operating the system as planned during extreme and nominal conditions and during the application of thermal gradients. It also accomplishes a bakeout of volatile materials, reducing the risk of degradation while in orbit due to contamination. Thermal vacuum tests should be carried out at pressures $<10^{-4}$ torr (1 torr = 1 mm Hg = 133.322 Pa). Most modern chambers have the capability of temperature excursion from -60°C to $+80^{\circ}\text{C}$ with typical vacuums of 10^{-7} to 10^{-8} torr. During the test, potential contamination of payloads is monitored by several methods, including temperature controlled quartz crystal microbalances (TCQM), cryogenic quartz crystal microbalances (CQCM), cold plates, and residual gas analyzers.

During test, the thermal environment may be supplied by one of the following methods:

1) Absorbed flux: The incident heat flux is produced by using heater panels or infrared lamps with the spectrum adjusted for the external coatings or electrical resistance heaters attached to the system's surfaces.

2) Incident flux: The intensity, spectrum, and angular distribution of the incident solar, albedo, and planetary heat flux are simulated.

3) Equivalent radiation sink temperature: The radiation sink (or surface) temperature of the system is simulated using infrared lamps and calorimeters with optical properties identical to that of the system.

4) Combination: The thermal environment is supplied by a combination of the preceding methods.

A mass estimate of the thermal control system of a spacecraft is around 5% of the dry spacecraft mass. A cost estimate of the thermal control system is around 5% of the total spacecraft cost.

12.8 Thermal Analysis Equations for Earth Satellites

A thermal analysis assumes that the heat flux input plus the heat flux output is equal to the stored heat, where in a steady-state analysis the latter is assumed constant. The heat flux balance for an element of an instrument or spacecraft in the vicinity of the Earth can be categorized as

$$\begin{aligned} \dot{Q}_{\text{Heat from the sun}} + \dot{Q}_{\text{IR heat from Earth}} + \dot{Q}_{\text{Albedo heat from Earth}} + \dot{Q}_{\text{Heat exchange with space}} \\ + \dot{Q}_{\text{Heat exchange with other surfaces}} + \dot{Q}_{\text{Heat dissipation}} + \dot{Q}_{\text{Heat stored}} = 0 \end{aligned} \quad (12.93)$$

The following formulas give the heat received by a flat plate and a sphere.

1) Heat received by a flat plate from the sun: From Eq. (12.82), it follows that

$$\dot{Q}_{\text{Heat from the sun}} = \alpha_s \dot{q}_{\text{sun}} A_p \sin^2 \eta_s \cos \lambda_s = \alpha_s S_e A_p \cos \lambda_s \quad (12.94)$$

2) Heat received by a spherical body from the sun: From Eqs. (12.77) and (12.79), it follows that

$$\dot{Q}_{\text{Heat from the sun}} = \frac{1}{2} \alpha_s \dot{q}_{\text{sun}} A_s (1 - \cos \eta_s) \approx \alpha_s S_e A_p \quad (12.95)$$

3) Heat received by a flat plate from the Earth infrared: From Eq. (12.82), it follows that

$$\dot{Q}_{\text{Infrared heat from Earth}} = \varepsilon_{\text{ir}} \dot{q}_{\text{Earth}} A_p \cos \lambda_e \sin^2 \eta_e \quad (12.96)$$

4) Heat received by a spherical body from the Earth infrared: From Eq. (12.77), it follows that

$$\begin{aligned} \dot{Q}_{\text{Infrared heat from Earth}} &= \frac{1}{2} \varepsilon_{\text{ir}} \dot{q}_{\text{Earth}} (1 - \cos \eta_e) A_s \\ &= 2 \varepsilon_{\text{ir}} \dot{q}_{\text{Earth}} (1 - \cos \eta_e) A_p \end{aligned} \quad (12.97)$$

5) Heat received by a rotating flat plate with spin axis pointing toward the Earth center from the Earth's albedo: From Eq. (12.88), it follows that

$$\dot{Q}_{\text{Albedo heat from Earth}} = \alpha_s \left(\frac{\dot{Q}_{a \rightarrow s}(h, \lambda, \theta_s)}{A_p} \right) \left(\frac{A_b}{0.34} \right) \left(\frac{S_e}{1343} \right) A_p \quad (12.98)$$

6) Heat received by a spherical body from the Earth albedo: From Eq. (12.85), it follows that

$$\dot{Q}_{\text{Albedo heat from Earth}} = \alpha_s \left(\frac{\dot{Q}_{a \rightarrow s}(h, \theta_s)}{A_p} \right) \left(\frac{A_b}{0.34} \right) \left(\frac{S_e}{1395} \right) A_p \quad (12.99)$$

7) Heat received from space: From Eq. (12.31), it follows that

$$\dot{Q}_{\text{Heat exchange with space}} = \varepsilon_{\text{ir}} \sigma F_{s, p \rightarrow \text{space}} A_s (T_{\text{space}}^4 - T_s^4) \approx -\varepsilon_{\text{ir}} \sigma A_s T_s^4 \quad (12.100)$$

8) Heat received by conduction with other surfaces by conduction: From Eq. (12.3), it follows that

$$\dot{Q}_{\text{Conduction heat exchange with other surfaces}} = - \sum_{j=1}^m k A_i \nabla T_s \quad (12.101)$$

9) Heat received on surface A_j from surface A_i : From Eq. (12.58), it follows that

$$\dot{Q}_{\text{Heat exchange between surfaces } i \text{ and } j} = \varepsilon_i \varepsilon_j \sigma A_i F_{i \rightarrow j} (T_i^4 - T_j^4) \quad (12.102)$$

10) Heat stored: From Eq. (12.6), it follows that

$$\dot{Q}_{\text{Heat stored}} = c_p m \frac{dT_s}{dt} \quad (12.103)$$

12.9 Steady-State Thermal Analysis

Examples of steady-state thermal analysis are given in the following.

Example 12.4

Problem: Determine the equations for the minimum and maximum steady-state temperatures of a thin plate insulated on the side facing away from the Earth whose normal is pointing toward the center of the Earth.

Solution: For the maximum temperature, the plate has the insulated side facing the sun with the bottom exposed to both the Earth infrared radiation and the Earth albedo. The steady-state heat balance is

$$\dot{Q}_{\text{Infrared heat from Earth}} + \dot{Q}_{\text{Albedo from Earth}} + \dot{Q}_{\text{Heat exchange with space}} = 0 \quad (12.104)$$

$$\begin{aligned} \varepsilon_{\text{ir}} \dot{q}_{\text{Earth}} A_p \cos \lambda_e \sin^2 \eta_e + \alpha_s \left(\frac{\dot{Q}_{a \rightarrow p}(h, \lambda, \theta_s)}{A_p} \right) \left(\frac{A_b}{0.34} \right) \\ \times \left(\frac{S_e}{1395} \right) A_p - \varepsilon_{\text{ir}} \sigma T_{\text{plate}}^4 A_p \approx 0 \end{aligned} \quad (12.105)$$

Since $\lambda_e = 0$, angle between normal to surface and direction to center of Earth, the maximum steady-state temperature is

$$T_{\text{plate, max}} = \left[\frac{\varepsilon_{\text{ir}} \dot{q}_{\text{Earth}} \sin^2 \eta_e + \alpha_s \left(\frac{\dot{Q}_{a \rightarrow p}(h, \lambda, \theta_s)}{A_p} \right) \left(\frac{A_b}{0.34} \right) \left(\frac{S_e}{1395} \right)}{\varepsilon_{\text{ir}} \sigma} \right]^{1/4} \quad (12.106)$$

The minimum temperature occurs when the Earth albedo is not present, $A_b = 0$, so that Eq. (12.106) gives

$$T_{\text{plate, min}} = \left[\frac{\dot{q}_{\text{Earth}} \sin^2 \eta}{\sigma} \right]^{1/4} \quad (12.107)$$

Example 12.5

Problem: Determine the equations for the steady-state temperature of a spherical spacecraft covered with solar cells with an efficiency of η_{sc} in orbit around the Earth.

Solution: For the maximum temperature, the spacecraft is exposed to the sun, Earth albedo, Earth infrared, and space radiations and dissipation of internal heat where

$$\dot{Q}_{\text{Heat from sun}} + \dot{Q}_{\text{Infrared heat from Earth}} + \dot{Q}_{\text{Albedo heat from Earth}} + \dot{Q}_{\text{Heat exchange with space}} + \dot{Q}_{\text{Heat dissipation}} = 0 \quad (12.108)$$

$$(1 - \eta_{sc})\alpha_s S_e A_p + \frac{1}{2} \varepsilon_{ir} \dot{q}_{\text{Earth}} (1 - \cos \eta_e) A_s + (1 - \eta_{sc})\alpha_s \left(\frac{\dot{Q}_{a \rightarrow s}(h, \lambda, \theta_s)}{A_p} \right) \times \left(\frac{A_b}{0.34} \right) \left(\frac{S_e}{1343} \right) A_p - \varepsilon_{ir} \sigma T_{\text{sphere}}^4 A_s + \dot{Q}_D \approx 0 \quad (12.109)$$

so that

$$T_{\text{sphere, max}} = \left[\frac{(1 - \eta_s)\alpha_s S_e A_p + 2\varepsilon_{ir} \dot{q}_{\text{Earth}} (1 - \cos \eta_e) A_p + (1 - \eta_s)\alpha_s \left(\frac{\dot{Q}_{a \rightarrow s}(h, \lambda, \theta_s)}{A_p} \right) \left(\frac{A_b}{0.34} \right) \left(\frac{S_e}{1343} \right) A_p + \dot{Q}_D}{4\varepsilon_{ir} \sigma A_p} \right]^{1/4} \quad (12.110)$$

The minimum steady-state temperature follows when the Earth albedo and the solar heat are absent as

$$T_{\text{sphere, min}} = \left[\frac{2\varepsilon_{ir} \dot{q}_{\text{Earth}} (1 - \cos \eta_e) A_p + \dot{Q}_D}{4\varepsilon_{ir} \sigma A_p} \right]^{1/4} \quad (12.111)$$

12.10 Transient Thermal Analysis

An example of transient thermal analysis is given in the following.

Example 12.6

Problem: Determine the transient temperatures of a spherical spacecraft in a circular orbit around the Earth with the following characteristics:

- A_b = albedo, 0.3
- $\dot{Q}_D$ = heat dissipated, 100 W
- m = mass, 200 kg
- A_s = surface area, $4\pi \text{ m}^2$
- h = satellite altitude, 1000 km
- c_p = specific heat average, $300 \text{ J kg}^{-1} \text{ K}^{-1}$
- $\dot{q}_{\text{Earth}}$ = heat flux density of surface of planetary body, 237 W m^{-2}

Solution: Conservation of heat flux gives that

$$\begin{aligned} \dot{Q}_{\text{Heat from the sun}} + \dot{Q}_{\text{IR heat from Earth}} + \dot{Q}_{\text{Albedo heat from Earth}} + \dot{Q}_{\text{Heat exchange with space}} \\ + \dot{Q}_{\text{Heat exchange with other surfaces}} + \dot{Q}_{\text{Heat dissipation}} + \dot{Q}_{\text{Heat stored}} = 0 \end{aligned} \quad (12.112)$$

that can be expressed as

$$\begin{aligned} \alpha_s S_e A_p + 2\epsilon_{\text{ir}} \dot{q}_{\text{Earth}} (1 - \cos \eta_e) A_p \\ + \alpha_s \left(\frac{\dot{Q}_{a \rightarrow s}(h, \lambda, \theta_s)}{A_p} \right) \left(\frac{A_b}{0.34} \right) \left(\frac{S_e}{1343} \right) A_p \\ - 4\epsilon_{\text{ir}} \sigma T_{\text{sphere}}^4 A_p + \dot{Q}_D \approx c_p m \frac{dT_{\text{sphere}}}{dt} \end{aligned} \quad (12.113)$$

Sample cases of the temperature variation while the sun is in the orbit plane so that there is a maximum time of eclipse are shown in Fig. 12.25. These were obtained by numerically integrating Eq. (12.113).

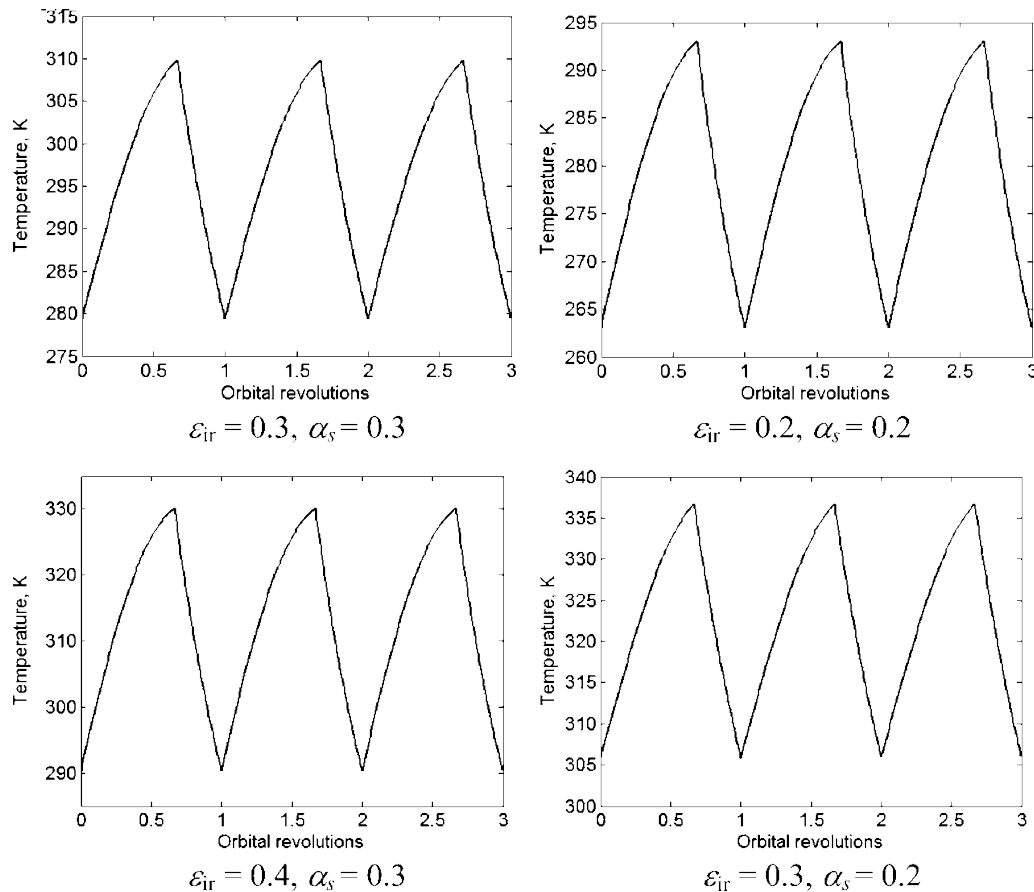


Fig. 12.25 Transient temperature examples.

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Problems

- 12.1** A hollow cylinder of length l insulated at the ends has an inside radius r_i , outside radius r_o , inside temperature T_i , and outside temperature T_o .
- Determine the equation for the radial temperature distribution.
 - Determine the radial heat flux density.
 - Determine the radial heat flux.
- 12.2** A hollow sphere has an internal radius r_i , outside radius r_o , internal temperature T_i , and outside temperature T_o .
- Determine the equation for the radial temperature distribution as a function of the radius r .
 - Determine the radial heat flux density.
 - Determine the radial heat flux.
- 12.3** Determine the temperatures at the node points 1, 2, 3, 4 in Fig. P12.3.

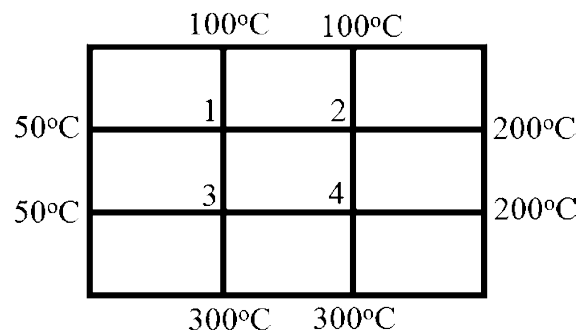


Fig. P12.3

- 12.4** A structure in a spacecraft, illustrated in Fig. P12.4, has $L = 25$ cm, $W = 30$ cm and is made of copper. The front face is attached to an instrument that has a temperature $T_H = 125^\circ\text{C}$ while the back face is exposed to a radiator at $T_C = 10^\circ\text{C}$. Determine the height H that is required to conduct 1660 W of heat at equilibrium to the radiator.